

# JESS Thermodynamic Database v8.9

C24

24-Jan-23

| Reaction No. 8820                         |    |       |        |                    |   |       |        |
|-------------------------------------------|----|-------|--------|--------------------|---|-------|--------|
| H+1 + AspArgValTyr-3 = H+1_AspArgValTyr-3 |    |       |        |                    |   |       |        |
| No                                        | t  | I-Str | Medium | Value (Dev)        | W | Tech. | Ref. # |
| 1                                         | 25 | 0.1   | KNO3   | lgK:10.27 (0.01SD) | 6 | MGL   | 1024   |

| Reaction No. 8821                               |    |       |        |                    |   |       |        |
|-------------------------------------------------|----|-------|--------|--------------------|---|-------|--------|
| 2<H+1> + AspArgValTyr-3 = H+1(2)_AspArgValTyr-3 |    |       |        |                    |   |       |        |
| No                                              | t  | I-Str | Medium | Value (Dev)        | W | Tech. | Ref. # |
| 1                                               | 25 | 0.1   | KNO3   | lgK:18.04 (0.02SD) | 4 | MGL   | 1024   |

| Reaction No. 8822                               |    |       |        |                    |   |       |        |
|-------------------------------------------------|----|-------|--------|--------------------|---|-------|--------|
| 3<H+1> + AspArgValTyr-3 = H+1(3)_AspArgValTyr-3 |    |       |        |                    |   |       |        |
| No                                              | t  | I-Str | Medium | Value (Dev)        | W | Tech. | Ref. # |
| 1                                               | 25 | 0.1   | KNO3   | lgK:21.87 (0.02SD) | 4 | MGL   | 1024   |

| Reaction No. 8823                               |    |       |        |                   |   |       |        |
|-------------------------------------------------|----|-------|--------|-------------------|---|-------|--------|
| 4<H+1> + AspArgValTyr-3 = H+1(4)_AspArgValTyr-3 |    |       |        |                   |   |       |        |
| No                                              | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
| 1                                               | 25 | 0.1   | KNO3   | lgK:25.0 (0.04SD) | 4 | MGL   | 1024   |

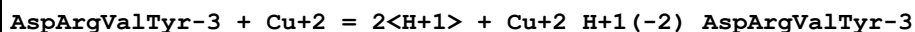
| Reaction No. 8837                                     |    |       |        |                    |   |       |        |
|-------------------------------------------------------|----|-------|--------|--------------------|---|-------|--------|
| H+1 + AspArgValTyr-3 + Cu+2 = Cu+2_H+1_AspArgValTyr-3 |    |       |        |                    |   |       |        |
| No                                                    | t  | I-Str | Medium | Value (Dev)        | W | Tech. | Ref. # |
| 1                                                     | 25 | 0.1   | KNO3   | lgK:16.34 (0.01SD) | 6 | MGL   | 1024   |

| Reaction No. 8838                           |    |       |        |                    |   |       |        |
|---------------------------------------------|----|-------|--------|--------------------|---|-------|--------|
| AspArgValTyr-3 + Cu+2 = Cu+2_AspArgValTyr-3 |    |       |        |                    |   |       |        |
| No                                          | t  | I-Str | Medium | Value (Dev)        | W | Tech. | Ref. # |
| 1                                           | 25 | 0.1   | KNO3   | lgK:10.14 (0.01SD) | 3 | MGL   | 1024   |

| Reaction No. 8839                                         |   |       |        |             |   |       |        |
|-----------------------------------------------------------|---|-------|--------|-------------|---|-------|--------|
| AspArgValTyr-3 + Cu+2 = H+1 + Cu+2_H+1(-1)_AspArgValTyr-3 |   |       |        |             |   |       |        |
| No                                                        | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |

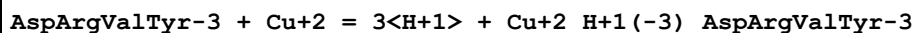
|   |    |     |      |                   |   |     |      |
|---|----|-----|------|-------------------|---|-----|------|
| 1 | 25 | 0.1 | KNO3 | lgK:2.34 (0.02SD) | 6 | MGL | 1024 |
|---|----|-----|------|-------------------|---|-----|------|

## Reaction No. 8840



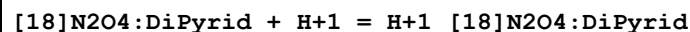
| No | t  | I-Str | Medium | Value (Dev)        | W | Tech. | Ref. # |
|----|----|-------|--------|--------------------|---|-------|--------|
| 1  | 25 | 0.1   | KNO3   | lgK:-7.13 (0.02SD) | 4 | MGL   | 1024   |

## Reaction No. 8841



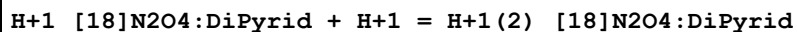
| No | t  | I-Str | Medium | Value (Dev)        | W | Tech. | Ref. # |
|----|----|-------|--------|--------------------|---|-------|--------|
| 1  | 25 | 0.1   | KNO3   | lgK:-17.9 (0.03SD) | 4 | MGL   | 1024   |

## Reaction No. 15134



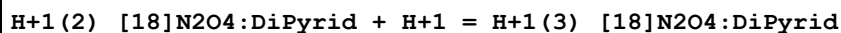
| No | t  | I-Str | Medium  | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|---------|-------------------|---|-------|--------|
| 1  | 25 | 0.1   | Unknown | lgK:7.44 (0.01SD) | 6 | MGL   | 1350   |

## Reaction No. 15135



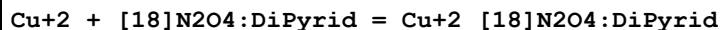
| No | t  | I-Str | Medium  | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|---------|-------------------|---|-------|--------|
| 1  | 25 | 0.1   | Unknown | lgK:6.26 (0.01SD) | 6 | MGL   | 1350   |

## Reaction No. 15136



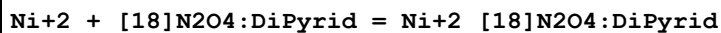
| No | t  | I-Str | Medium  | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|---------|------------------|---|-------|--------|
| 1  | 25 | 0.1   | Unknown | lgK:1.4 (0.03SD) | 3 | MGL   | 1350   |

## Reaction No. 15137



| No | t  | I-Str | Medium  | Value (Dev)        | W | Tech. | Ref. # |
|----|----|-------|---------|--------------------|---|-------|--------|
| 1  | 25 | 0.1   | Unknown | lgK:13.55 (0.02SD) | 6 | MGL   | 1350   |

## Reaction No. 15138



| No | t  | I-Str | Medium  | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|---------|-------------------|---|-------|--------|
| 1  | 25 | 0.1   | Unknown | lgK:8.80 (0.02SD) | 4 | MGL   | 1350   |

## Reaction No. 15139

| Zn+2 + [18]N2O4:DiPyrid = Zn+2_[18]N2O4:DiPyrid |    |       |         |                  |   |       |        |
|-------------------------------------------------|----|-------|---------|------------------|---|-------|--------|
| No                                              | t  | I-Str | Medium  | Value (Dev)      | W | Tech. | Ref. # |
| 1                                               | 25 | 0.1   | Unknown | lgK:6.96(0.01SD) | 4 | MGL   | 1350   |

| Reaction No. 15140                              |    |       |         |                   |   |       |        |
|-------------------------------------------------|----|-------|---------|-------------------|---|-------|--------|
| Cd+2 + [18]N2O4:DiPyrid = Cd+2_[18]N2O4:DiPyrid |    |       |         |                   |   |       |        |
| No                                              | t  | I-Str | Medium  | Value (Dev)       | W | Tech. | Ref. # |
| 1                                               | 25 | 0.1   | Unknown | lgK:10.96(0.02SD) | 4 | MGL   | 1350   |

| Reaction No. 15141                              |    |       |         |                  |   |       |        |
|-------------------------------------------------|----|-------|---------|------------------|---|-------|--------|
| Ca+2 + [18]N2O4:DiPyrid = Ca+2_[18]N2O4:DiPyrid |    |       |         |                  |   |       |        |
| No                                              | t  | I-Str | Medium  | Value (Dev)      | W | Tech. | Ref. # |
| 1                                               | 25 | 0.1   | Unknown | lgK:3.63(0.02SD) | 6 | MGL   | 1350   |

| Reaction No. 15142                              |    |       |         |                 |   |       |        |
|-------------------------------------------------|----|-------|---------|-----------------|---|-------|--------|
| La+3 + [18]N2O4:DiPyrid = La+3_[18]N2O4:DiPyrid |    |       |         |                 |   |       |        |
| No                                              | t  | I-Str | Medium  | Value (Dev)     | W | Tech. | Ref. # |
| 1                                               | 25 | 0.1   | Unknown | lgK:3.5(0.05SD) | 4 | MGL   | 1350   |

| Reaction No. 15143                              |    |       |         |                  |   |       |        |
|-------------------------------------------------|----|-------|---------|------------------|---|-------|--------|
| Sr+2 + [18]N2O4:DiPyrid = Sr+2_[18]N2O4:DiPyrid |    |       |         |                  |   |       |        |
| No                                              | t  | I-Str | Medium  | Value (Dev)      | W | Tech. | Ref. # |
| 1                                               | 25 | 0.1   | Unknown | lgK:4.87(0.01SD) | 4 | MGL   | 1350   |

| Reaction No. 15144                              |    |       |         |                   |   |       |        |
|-------------------------------------------------|----|-------|---------|-------------------|---|-------|--------|
| Pb+2 + [18]N2O4:DiPyrid = Pb+2_[18]N2O4:DiPyrid |    |       |         |                   |   |       |        |
| No                                              | t  | I-Str | Medium  | Value (Dev)       | W | Tech. | Ref. # |
| 1                                               | 25 | 0.1   | Unknown | lgK:11.67(0.01SD) | 6 | MGL   | 1350   |

| Reaction No. 15145                              |    |       |         |                  |   |       |        |
|-------------------------------------------------|----|-------|---------|------------------|---|-------|--------|
| Ba+2 + [18]N2O4:DiPyrid = Ba+2_[18]N2O4:DiPyrid |    |       |         |                  |   |       |        |
| No                                              | t  | I-Str | Medium  | Value (Dev)      | W | Tech. | Ref. # |
| 1                                               | 25 | 0.1   | Unknown | lgK:4.99(0.01SD) | 4 | MGL   | 1350   |

| Reaction No. 15201          |    |       |        |                 |   |       |        |
|-----------------------------|----|-------|--------|-----------------|---|-------|--------|
| H+1 + [36]N12 = H+1_[36]N12 |    |       |        |                 |   |       |        |
| No                          | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                           | 25 | 0.15  | NaClO4 | lgK:9.7(0.05SD) | 3 | MGL   | 1341   |

|   |    |      |                                  |                   |   |     |      |
|---|----|------|----------------------------------|-------------------|---|-----|------|
| 2 | 25 | 0.15 | DMSO:Water<br>80:20Vol<br>NaClO4 | lgK:10.23(0.01SD) | 5 | MGL | 2548 |
|---|----|------|----------------------------------|-------------------|---|-----|------|

| Reaction No. 15202                       |    |       |        |                 |   |       |        |
|------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| $H+1 + H+1_{[36]N12} = H+1(2)_{[36]N12}$ |    |       |        |                 |   |       |        |
| No                                       | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                        | 25 | 0.15  | NaClO4 | lgK:9.7(0.05SD) | 3 | MGL   | 1341   |

| Reaction No. 15203                          |    |       |        |                 |   |       |        |
|---------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| $H+1(2)_{[36]N12} + H+1 = H+1(3)_{[36]N12}$ |    |       |        |                 |   |       |        |
| No                                          | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                           | 25 | 0.15  | NaClO4 | lgK:8.9(0.05SD) | 4 | MGL   | 1341   |

| Reaction No. 15204                          |    |       |        |                 |   |       |        |
|---------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| $H+1(3)_{[36]N12} + H+1 = H+1(4)_{[36]N12}$ |    |       |        |                 |   |       |        |
| No                                          | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                           | 25 | 0.15  | NaClO4 | lgK:9.0(0.05SD) | 4 | MGL   | 1341   |

| Reaction No. 15205                          |    |       |        |                 |   |       |        |
|---------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| $H+1(4)_{[36]N12} + H+1 = H+1(5)_{[36]N12}$ |    |       |        |                 |   |       |        |
| No                                          | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                           | 25 | 0.15  | NaClO4 | lgK:8.1(0.05SD) | 4 | MGL   | 1341   |

| Reaction No. 15206                          |    |       |        |                 |   |       |        |
|---------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| $H+1(5)_{[36]N12} + H+1 = H+1(6)_{[36]N12}$ |    |       |        |                 |   |       |        |
| No                                          | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                           | 25 | 0.15  | NaClO4 | lgK:7.8(0.05SD) | 4 | MGL   | 1341   |

| Reaction No. 15207                          |    |       |        |                 |   |       |        |
|---------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| $H+1(6)_{[36]N12} + H+1 = H+1(7)_{[36]N12}$ |    |       |        |                 |   |       |        |
| No                                          | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                           | 25 | 0.15  | NaClO4 | lgK:5.7(0.05SD) | 4 | MGL   | 1341   |

| Reaction No. 15208                          |    |       |        |                 |   |       |        |
|---------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| $H+1(7)_{[36]N12} + H+1 = H+1(8)_{[36]N12}$ |    |       |        |                 |   |       |        |
| No                                          | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                           | 25 | 0.15  | NaClO4 | lgK:4.3(0.04SD) | 4 | MGL   | 1341   |

| Reaction No. 15209                                                             |    |       |        |                 |   |       |        |
|--------------------------------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| $\text{H+1(8)}_{[36]\text{N12}} + \text{H+1} = \text{H+1(9)}_{[36]\text{N12}}$ |    |       |        |                 |   |       |        |
| No                                                                             | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                              | 25 | 0.15  | NaClO4 | lgK:3.6(0.04SD) | 4 | MGL   | 1341   |

| Reaction No. 15210                                                              |    |       |        |                 |   |       |        |
|---------------------------------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| $\text{H+1(9)}_{[36]\text{N12}} + \text{H+1} = \text{H+1(10)}_{[36]\text{N12}}$ |    |       |        |                 |   |       |        |
| No                                                                              | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                               | 25 | 0.15  | NaClO4 | lgK:2.6(0.04SD) | 4 | MGL   | 1341   |

| Reaction No. 15211                                                               |    |       |        |                |   |       |        |
|----------------------------------------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| $\text{H+1(10)}_{[36]\text{N12}} + \text{H+1} = \text{H+1(11)}_{[36]\text{N12}}$ |    |       |        |                |   |       |        |
| No                                                                               | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                                | 25 | 0.15  | NaClO4 | lgK:2.3(0.1SD) | 3 | MGL   | 1341   |

| Reaction No. 15212                                                               |    |       |        |              |   |       |        |
|----------------------------------------------------------------------------------|----|-------|--------|--------------|---|-------|--------|
| $\text{H+1(11)}_{[36]\text{N12}} + \text{H+1} = \text{H+1(12)}_{[36]\text{N12}}$ |    |       |        |              |   |       |        |
| No                                                                               | t  | I-Str | Medium | Value (Dev)  | W | Tech. | Ref. # |
| 1                                                                                | 25 | 0.15  | NaClO4 | lgK:1.0(2SF) | 6 | MGL   | 1341   |

| Reaction No. 15213                                                  |    |       |        |                  |   |       |        |
|---------------------------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| $2\text{<Cu+2>} + [36]\text{N12} = \text{Cu+2(2)}_{[36]\text{N12}}$ |    |       |        |                  |   |       |        |
| No                                                                  | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                                   | 25 | 0.15  | NaClO4 | lgK:39.3(0.07SD) | 3 | MGL   | 1341   |

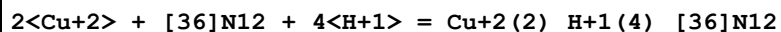
| Reaction No. 15214                                                                            |    |       |        |                  |   |       |        |
|-----------------------------------------------------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| $2\text{<Cu+2>} + [36]\text{N12} + \text{H+1} = \text{Cu+2(2)}_{\text{H+1}}_{[36]\text{N12}}$ |    |       |        |                  |   |       |        |
| No                                                                                            | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                                                             | 25 | 0.15  | NaClO4 | lgK:47.1(0.07SD) | 4 | MGL   | 1341   |

| Reaction No. 15215                                                                                  |    |       |        |                  |   |       |        |
|-----------------------------------------------------------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| $2\text{<Cu+2>} + [36]\text{N12} + 2\text{<H+1>} = \text{Cu+2(2)}_{\text{H+1(2)}}_{[36]\text{N12}}$ |    |       |        |                  |   |       |        |
| No                                                                                                  | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                                                                   | 25 | 0.15  | NaClO4 | lgK:54.0(0.08SD) | 4 | MGL   | 1341   |

| Reaction No. 15216                                                                                  |   |       |        |             |   |       |        |
|-----------------------------------------------------------------------------------------------------|---|-------|--------|-------------|---|-------|--------|
| $2\text{<Cu+2>} + [36]\text{N12} + 3\text{<H+1>} = \text{Cu+2(2)}_{\text{H+1(3)}}_{[36]\text{N12}}$ |   |       |        |             |   |       |        |
| No                                                                                                  | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |

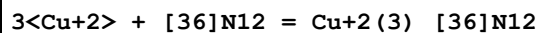
|   |    |      |        |                  |   |     |      |
|---|----|------|--------|------------------|---|-----|------|
| 1 | 25 | 0.15 | NaClO4 | lgK:58.1(0.08SD) | 4 | MGL | 1341 |
|---|----|------|--------|------------------|---|-----|------|

## Reaction No. 15217



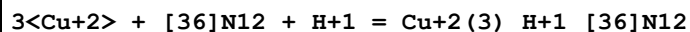
| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 0.15  | NaClO4 | lgK:61.4(0.08SD) | 4 | MGL   | 1341   |

## Reaction No. 15218



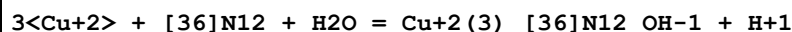
| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 0.15  | NaClO4 | lgK:51.4(0.07SD) | 4 | MGL   | 1341   |

## Reaction No. 15219



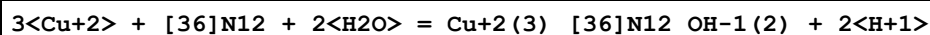
| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 0.15  | NaClO4 | lgK:55.4(0.07SD) | 4 | MGL   | 1341   |

## Reaction No. 15220



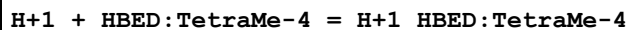
| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 0.15  | NaClO4 | lgK:42.5(0.09SD) | 4 | MGL   | 1341   |

## Reaction No. 15221



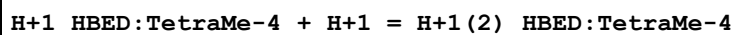
| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 0.15  | NaClO4 | lgK:31.6(0.09SD) | 4 | MGL   | 1341   |

## Reaction No. 16420



| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | KCl    | lgK:13.12(4SF) | 6 | MGL   | 1358   |

## Reaction No. 16421



| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | KCl    | lgK:12.99(4SF) | 6 | MGL   | 1358   |

## Reaction No. 16422

| H+1(2)_HBED:TetraMe-4 + H+1 = H+1(3)_HBED:TetraMe-4 |    |       |        |                |   |       |        |
|-----------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| No                                                  | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                   | 25 | 0.1   | KCl    | lgK:7.94 (3SF) | 6 | MGL   | 1358   |

| Reaction No. 16423                                  |    |       |        |                |   |       |        |
|-----------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| H+1(3)_HBED:TetraMe-4 + H+1 = H+1(4)_HBED:TetraMe-4 |    |       |        |                |   |       |        |
| No                                                  | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                   | 25 | 0.1   | KCl    | lgK:4.09 (3SF) | 6 | MGL   | 1358   |

| Reaction No. 16424                                  |    |       |        |               |   |       |        |
|-----------------------------------------------------|----|-------|--------|---------------|---|-------|--------|
| H+1(4)_HBED:TetraMe-4 + H+1 = H+1(5)_HBED:TetraMe-4 |    |       |        |               |   |       |        |
| No                                                  | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                   | 25 | 0.1   | KCl    | lgK:1.4 (2SF) | 4 | MGL   | 1358   |

| Reaction No. 16425                          |    |       |        |                 |   |       |        |
|---------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| Ga+3 + HBED:TetraMe-4 = Ga+3_HBED:TetraMe-4 |    |       |        |                 |   |       |        |
| No                                          | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                           | 25 | 0.1   | KCl    | lgK:34. (0.3SD) | 3 | MGL   | 1358   |

| Reaction No. 16426                          |    |       |        |                 |   |       |        |
|---------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| Fe+3 + HBED:TetraMe-4 = Fe+3_HBED:TetraMe-4 |    |       |        |                 |   |       |        |
| No                                          | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                           | 25 | 0.1   | KCl    | lgK:37.41 (4SF) | 6 | MGL   | 1358   |

| Reaction No. 16427                          |    |       |        |                 |   |       |        |
|---------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| In+3 + HBED:TetraMe-4 = In+3_HBED:TetraMe-4 |    |       |        |                 |   |       |        |
| No                                          | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                           | 25 | 0.1   | KCl    | lgK:30.72 (4SF) | 6 | MGL   | 1358   |

| Reaction No. 16457                      |    |       |        |                   |   |       |        |
|-----------------------------------------|----|-------|--------|-------------------|---|-------|--------|
| H+1 + HBPLED:Me*4-3 = H+1_HBPLED:Me*4-3 |    |       |        |                   |   |       |        |
| No                                      | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
| 1                                       | 25 | 0.1   | KCl    | lgK:12.1 (0.03SD) | 4 | MGL   | 1358   |

| Reaction No. 16458                             |    |       |        |                   |   |       |        |
|------------------------------------------------|----|-------|--------|-------------------|---|-------|--------|
| H+1_HBPLED:Me*4-3 + H+1 = H+1(2)_HBPLED:Me*4-3 |    |       |        |                   |   |       |        |
| No                                             | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
| 1                                              | 25 | 0.1   | KCl    | lgK:9.72 (0.01SD) | 6 | MGL   | 1358   |

| Reaction No. 16459                                                                         |    |       |        |                  |   |       |        |
|--------------------------------------------------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| $\text{H+1(2)}_{\text{HBPLED:Me*4-3}} + \text{H+1} = \text{H+1(3)}_{\text{HBPLED:Me*4-3}}$ |    |       |        |                  |   |       |        |
| No                                                                                         | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                                                          | 25 | 0.1   | KCl    | lgK:5.99(0.01SD) | 4 | MGL   | 1358   |

| Reaction No. 16460                                                                         |    |       |        |                  |   |       |        |
|--------------------------------------------------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| $\text{H+1(3)}_{\text{HBPLED:Me*4-3}} + \text{H+1} = \text{H+1(4)}_{\text{HBPLED:Me*4-3}}$ |    |       |        |                  |   |       |        |
| No                                                                                         | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                                                          | 25 | 0.1   | KCl    | lgK:2.96(0.01SD) | 4 | MGL   | 1358   |

| Reaction No. 16461                                                                         |    |       |        |                |   |       |        |
|--------------------------------------------------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| $\text{H+1(4)}_{\text{HBPLED:Me*4-3}} + \text{H+1} = \text{H+1(5)}_{\text{HBPLED:Me*4-3}}$ |    |       |        |                |   |       |        |
| No                                                                                         | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                                          | 25 | 0.1   | KCl    | lgK:1.1(0.1SD) | 3 | MGL   | 1358   |

| Reaction No. 16462                                                        |    |       |        |                  |   |       |        |
|---------------------------------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| $\text{Ga+3} + \text{HBPLED:Me*4-3} = \text{Ga+3}_{\text{HBPLED:Me*4-3}}$ |    |       |        |                  |   |       |        |
| No                                                                        | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                                         | 25 | 0.1   | KCl    | lgK:31.9(0.08SD) | 4 | MGL   | 1358   |

| Reaction No. 16463                                                        |    |       |        |                  |   |       |        |
|---------------------------------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| $\text{Fe+3} + \text{HBPLED:Me*4-3} = \text{Fe+3}_{\text{HBPLED:Me*4-3}}$ |    |       |        |                  |   |       |        |
| No                                                                        | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                                         | 25 | 0.1   | KCl    | lgK:33.0(0.03SD) | 4 | MGL   | 1358   |

| Reaction No. 16464                                                        |    |       |        |                  |   |       |        |
|---------------------------------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| $\text{In+3} + \text{HBPLED:Me*4-3} = \text{In+3}_{\text{HBPLED:Me*4-3}}$ |    |       |        |                  |   |       |        |
| No                                                                        | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                                         | 25 | 0.1   | KCl    | lgK:27.8(0.03SD) | 4 | MGL   | 1358   |

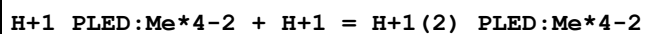
| Reaction No. 16631                                                          |    |       |        |                  |   |       |        |
|-----------------------------------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| $\text{Gd+3} + \text{HBED:TetraMe-4} = \text{Gd+3}_{\text{HBED:TetraMe-4}}$ |    |       |        |                  |   |       |        |
| No                                                                          | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                                           | 25 | 0.1   | KCl    | lgK:20.3(0.06SD) | 3 | MGL   | 1200   |

| Reaction No. 16632                                                  |  |  |  |  |  |  |  |
|---------------------------------------------------------------------|--|--|--|--|--|--|--|
| $\text{H+1} + \text{PLED:Me*4-2} = \text{H+1}_{\text{PLED:Me*4-2}}$ |  |  |  |  |  |  |  |



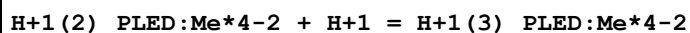
| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 0.1   | KCl    | lgK:8.95(0.01SD) | 4 | MGL   | 1358   |

## Reaction No. 16633



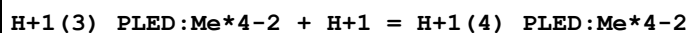
| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 0.1   | KCl    | lgK:6.85(0.01SD) | 6 | MGL   | 1358   |

## Reaction No. 16634



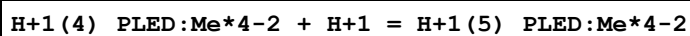
| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 0.1   | KCl    | lgK:3.30(0.01SD) | 6 | MGL   | 1358   |

## Reaction No. 16635



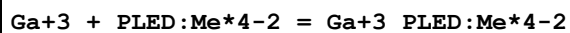
| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 0.1   | KCl    | lgK:2.24(0.01SD) | 6 | MGL   | 1358   |

## Reaction No. 16636



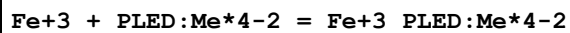
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | KCl    | lgK:1.3(0.1SD) | 3 | MGL   | 1358   |

## Reaction No. 16637



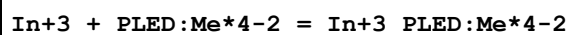
| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.1   | KCl    | lgK:27.3(0.2SD) | 3 | MGL   | 1358   |

## Reaction No. 16638



| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.1   | KCl    | lgK:27.2(0.2SD) | 3 | MGL   | 1358   |

## Reaction No. 16639



| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 0.1   | KCl    | lgK:21.5(0.03SD) | 6 | MGL   | 1358   |

| Reaction No. 16649        |    |       |        |                  |   |       |        |
|---------------------------|----|-------|--------|------------------|---|-------|--------|
| H+1 + HBMA-4 = H+1_HBMA-4 |    |       |        |                  |   |       |        |
| No                        | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                         | 25 | 0.1   | KCl    | lgK:12.9(0.03SD) | 4 | MGL   | 1200   |

| Reaction No. 16650               |    |       |        |                   |   |       |        |
|----------------------------------|----|-------|--------|-------------------|---|-------|--------|
| H+1_HBMA-4 + H+1 = H+1(2)_HBMA-4 |    |       |        |                   |   |       |        |
| No                               | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
| 1                                | 25 | 0.1   | KCl    | lgK:11.97(0.02SD) | 4 | MGL   | 1200   |

| Reaction No. 16651                  |    |       |        |                  |   |       |        |
|-------------------------------------|----|-------|--------|------------------|---|-------|--------|
| H+1(2)_HBMA-4 + H+1 = H+1(3)_HBMA-4 |    |       |        |                  |   |       |        |
| No                                  | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                   | 25 | 0.1   | KCl    | lgK:7.55(0.01SD) | 6 | MGL   | 1200   |

| Reaction No. 16652                  |    |       |        |                  |   |       |        |
|-------------------------------------|----|-------|--------|------------------|---|-------|--------|
| H+1(3)_HBMA-4 + H+1 = H+1(4)_HBMA-4 |    |       |        |                  |   |       |        |
| No                                  | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                   | 25 | 0.1   | KCl    | lgK:4.04(0.01SD) | 4 | MGL   | 1200   |

| Reaction No. 16653                  |    |       |        |                  |   |       |        |
|-------------------------------------|----|-------|--------|------------------|---|-------|--------|
| H+1(4)_HBMA-4 + H+1 = H+1(5)_HBMA-4 |    |       |        |                  |   |       |        |
| No                                  | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                   | 25 | 0.1   | KCl    | lgK:1.56(0.01SD) | 6 | MGL   | 1200   |

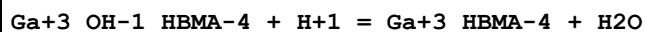
| Reaction No. 16654          |    |       |        |                  |   |       |        |
|-----------------------------|----|-------|--------|------------------|---|-------|--------|
| Fe+3 + HBMA-4 = Fe+3_HBMA-4 |    |       |        |                  |   |       |        |
| No                          | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                           | 25 | 0.1   | KCl    | lgK:31.2(0.06SD) | 4 | MGL   | 1200   |

| Reaction No. 16655                  |    |       |        |                  |   |       |        |
|-------------------------------------|----|-------|--------|------------------|---|-------|--------|
| Fe+3_HBMA-4 + H+1 = Fe+3_H+1_HBMA-4 |    |       |        |                  |   |       |        |
| No                                  | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                   | 25 | 0.1   | KCl    | lgK:3.43(0.01SD) | 6 | MGL   | 1200   |

| Reaction No. 16656          |   |       |        |             |   |       |        |
|-----------------------------|---|-------|--------|-------------|---|-------|--------|
| Ga+3 + HBMA-4 = Ga+3_HBMA-4 |   |       |        |             |   |       |        |
| No                          | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |

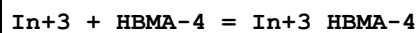
|   |    |     |     |                |   |     |      |
|---|----|-----|-----|----------------|---|-----|------|
| 1 | 25 | 0.1 | KCl | lgK:31.(0.3SD) | 3 | MGL | 1200 |
|---|----|-----|-----|----------------|---|-----|------|

## Reaction No. 16657



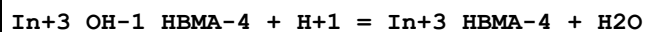
| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 0.1   | KCl    | lgK:6.71(0.01SD) | 6 | MGL   | 1200   |

## Reaction No. 16658



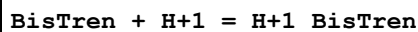
| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 0.1   | KCl    | lgK:26.3(0.06SD) | 4 | MGL   | 1200   |

## Reaction No. 16659



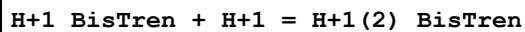
| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 0.1   | KCl    | lgK:8.37(0.01SD) | 6 | MGL   | 1200   |

## Reaction No. 17635



| No | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|--------|-------------------|---|-------|--------|
| 1  | 25 | 0.1   | KCl    | lgK:9.89(3SF)     | 6 | MGL   | 1361   |
| 2  | 25 | 0.1   | KCl    | lgK:9.88(0.003SD) | 6 | MGL   | 2459   |
| 3  | 25 | 0.1   | KNO3   | lgK:9.93(3SF)     | 6 | MGL   | 1361   |
| 4  | 25 | 0.1   | NaClO4 | lgK:9.89(3SF)     | 6 | MGL   | 1361   |
| 5  | 25 | 0.1   | NaClO4 | lgK:9.99(3SF)     | 6 | MGL   | 2908   |
| 6  | 25 | 1     | NaBr   | lgK:10.290(5SF)   | 6 | MGL   | 4358   |
| 7  | 25 | 1     | NaCl   | lgK:10.136(5SF)   | 6 | MGL   | 4358   |
| 8  | 25 | 1     | NaClO4 | lgK:10.384(5SF)   | 6 | MGL   | 4358   |
| 9  | 25 | 1     | NaI    | lgK:10.262(5SF)   | 5 | MGL   | 4358   |

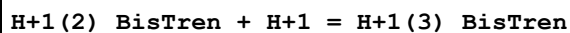
## Reaction No. 17636



| No | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|--------|-------------------|---|-------|--------|
| 1  | 25 | 0.1   | KCl    | lgK:9.17(3SF)     | 6 | MGL   | 1361   |
| 2  | 25 | 0.1   | KCl    | lgK:9.20(0.003SD) | 6 | MGL   | 2459   |
| 3  | 25 | 0.1   | KNO3   | lgK:9.31(3SF)     | 6 | MGL   | 1361   |
| 4  | 25 | 0.1   | NaClO4 | lgK:9.23(3SF)     | 6 | MGL   | 1361   |

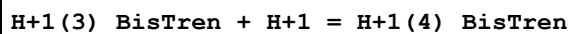
|   |    |     |                    |                 |   |     |      |
|---|----|-----|--------------------|-----------------|---|-----|------|
| 5 | 25 | 0.1 | NaClO <sub>4</sub> | lgK:9.02 (3SF)  | 6 | MGL | 2908 |
| 6 | 25 | 1   | NaBr               | lgK:9.751 (4SF) | 6 | MGL | 4358 |
| 7 | 25 | 1   | NaCl               | lgK:9.691 (4SF) | 6 | MGL | 4358 |
| 8 | 25 | 1   | NaClO <sub>4</sub> | lgK:9.815 (4SF) | 6 | MGL | 4358 |
| 9 | 25 | 1   | NaI                | lgK:9.778 (4SF) | 5 | MGL | 4358 |

## Reaction No. 17637



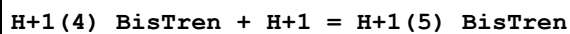
| No | t  | I-Str | Medium             | Value (Dev)        | W | Tech. | Ref. # |
|----|----|-------|--------------------|--------------------|---|-------|--------|
| 1  | 25 | 0.1   | KCl                | lgK:8.26 (3SF)     | 6 | MGL   | 1361   |
| 2  | 25 | 0.1   | KCl                | lgK:8.29 (0.003SD) | 6 | MGL   | 2459   |
| 3  | 25 | 0.1   | KNO <sub>3</sub>   | lgK:8.55 (3SF)     | 6 | MGL   | 1361   |
| 4  | 25 | 0.1   | NaClO <sub>4</sub> | lgK:8.29 (3SF)     | 6 | MGL   | 1361   |
| 5  | 25 | 0.1   | NaClO <sub>4</sub> | lgK:7.98 (3SF)     | 6 | MGL   | 2908   |
| 6  | 25 | 1     | NaBr               | lgK:8.978 (4SF)    | 6 | MGL   | 4358   |
| 7  | 25 | 1     | NaCl               | lgK:8.894 (4SF)    | 6 | MGL   | 4358   |
| 8  | 25 | 1     | NaClO <sub>4</sub> | lgK:9.010 (4SF)    | 6 | MGL   | 4358   |
| 9  | 25 | 1     | NaI                | lgK:8.958 (4SF)    | 5 | MGL   | 4358   |

## Reaction No. 17638



| No | t  | I-Str | Medium             | Value (Dev)        | W | Tech. | Ref. # |
|----|----|-------|--------------------|--------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution      | lgK:7.2 (2SF)      | 1 | EDH   |        |
| 2  | 25 | 0.1   | KCl                | lgK:7.89 (3SF)     | 6 | MGL   | 1361   |
| 3  | 25 | 0.1   | KCl                | lgK:7.84 (0.003SD) | 6 | MGL   | 2459   |
| 4  | 25 | 0.1   | KNO <sub>3</sub>   | lgK:7.91 (3SF)     | 6 | MGL   | 1361   |
| 5  | 25 | 0.1   | NaClO <sub>4</sub> | lgK:7.65 (3SF)     | 0 | MGL   | 1361   |
| 6  | 25 | 0.1   | NaClO <sub>4</sub> | lgK:7.20 (3SF)     | 0 | MGL   | 2908   |
| 7  | 25 | 1     | NaBr               | lgK:8.589 (4SF)    | 6 | MGL   | 4358   |
| 8  | 25 | 1     | NaCl               | lgK:8.653 (4SF)    | 6 | MGL   | 4358   |
| 9  | 25 | 1     | NaClO <sub>4</sub> | lgK:8.404 (4SF)    | 6 | MGL   | 4358   |
| 10 | 25 | 1     | NaI                | lgK:8.427 (4SF)    | 5 | MGL   | 4358   |

## Reaction No. 17639



| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | KCl    | lgK:7.28 (3SF) | 5 | MGL   | 1361   |

|                                            |    |     |        |                   |   |     |      |
|--------------------------------------------|----|-----|--------|-------------------|---|-----|------|
| 2                                          | 25 | 0.1 | KCl    | lgK:7.29(0.003SD) | 5 | MGL | 2459 |
| 3                                          | 25 | 0.1 | KNO3   | lgK:7.32(3SF)     | 5 | MGL | 1361 |
| 4                                          | 25 | 0.1 | NaClO4 | lgK:6.64(3SF)     | 0 | MGL | 1361 |
| 5                                          | 25 | 0.1 | NaClO4 | lgK:6.40(3SF)     | 0 | MGL | 2908 |
| Note (5): Low wgt for internal consistency |    |     |        |                   |   |     |      |
| 6                                          | 25 | 1   | NaBr   | lgK:7.930(4SF)    | 5 | MGL | 4358 |
| 7                                          | 25 | 1   | NaCl   | lgK:8.065(4SF)    | 5 | MGL | 4358 |
| 8                                          | 25 | 1   | NaClO4 | lgK:7.444(4SF)    | 5 | MGL | 4358 |
| 9                                          | 25 | 1   | NaI    | lgK:7.596(4SF)    | 4 | MGL | 4358 |

| Reaction No. 17640                         |    |       |        |                   |   |       |        |
|--------------------------------------------|----|-------|--------|-------------------|---|-------|--------|
| H+1(5)_BisTren + H+1 = H+1(6)_BisTren      |    |       |        |                   |   |       |        |
| No                                         | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
| 1                                          | 25 | 0.1   | KCl    | lgK:6.60(3SF)     | 5 | MGL   | 1361   |
| 2                                          | 25 | 0.1   | KCl    | lgK:6.55(0.003SD) | 5 | MGL   | 2459   |
| 3                                          | 25 | 0.1   | KNO3   | lgK:6.63(3SF)     | 5 | MGL   | 1361   |
| 4                                          | 25 | 0.1   | NaClO4 | lgK:6.01(3SF)     | 0 | MGL   | 1361   |
| Note (4): Low wgt for internal consistency |    |       |        |                   |   |       |        |
| 5                                          | 25 | 0.1   | NaClO4 | lgK:5.67(3SF)     | 0 | MGL   | 2908   |
| 6                                          | 25 | 1     | NaBr   | lgK:7.216(4SF)    | 5 | MGL   | 4358   |
| 7                                          | 25 | 1     | NaCl   | lgK:7.454(4SF)    | 5 | MGL   | 4358   |
| 8                                          | 25 | 1     | NaClO4 | lgK:6.954(4SF)    | 5 | MGL   | 4358   |
| 9                                          | 25 | 1     | NaI    | lgK:6.785(4SF)    | 4 | MGL   | 4358   |

| Reaction No. 17641            |    |       |        |                |   |       |        |
|-------------------------------|----|-------|--------|----------------|---|-------|--------|
| BisTren + Cu+2 = Cu+2_BisTren |    |       |        |                |   |       |        |
| No                            | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                             | 25 | 0.025 | NaF    | lgK:18.30(4SF) | 0 | MGL   | 4358   |
| 2                             | 25 | 0.1   | KCl    | lgK:16.54(4SF) | 0 | MGL   | 1361   |
| 3                             | 25 | 0.1   | NaClO4 | lgK:17.36(4SF) | 4 | MGL   | 1545   |
| 4                             | 25 | 0.1   | NaClO4 | lgK:17.59(4SF) | 4 | MGL   | 4379   |
| 5                             | 25 | 0.5   | NaCl   | lgK:17.85(4SF) | 4 | MGL   | 4358   |
| 6                             | 25 | 0.5   | NaClO4 | lgK:17.85(4SF) | 4 | MGL   | 4358   |
| 7                             | 25 | 1     | NaBr   | lgK:18.15(4SF) | 4 | MGL   | 4358   |
| 8                             | 25 | 1     | NaCl   | lgK:17.34(4SF) | 0 | MGL   | 4358   |
| 9                             | 25 | 1     | NaClO4 | lgK:18.65(4SF) | 0 | MGL   | 4358   |

| Reaction No. 17642                                                |    |       |                    |                             |   |       |        |
|-------------------------------------------------------------------|----|-------|--------------------|-----------------------------|---|-------|--------|
| $2\text{Cu}^{2+} + \text{BisTren} = \text{Cu}_2(2)\text{BisTren}$ |    |       |                    |                             |   |       |        |
| No                                                                | t  | I-Str | Medium             | Value (Dev)                 | W | Tech. | Ref. # |
| 1                                                                 | 25 | 0.025 | NaF                | $\lg K: 32.9 (3\text{SF})$  | 0 | MGL   | 4358   |
| Note (1): Low wgt for internal consistency                        |    |       |                    |                             |   |       |        |
| 2                                                                 | 25 | 0.1   | KCl                | $\lg K: 29.05 (4\text{SF})$ | 0 | MGL   | 1361   |
| 3                                                                 | 25 | 0.1   | NaClO <sub>4</sub> | $\lg K: 27.32 (4\text{SF})$ | 0 | MGL   | 1361   |
| Note (3): Low wgt for internal consistency                        |    |       |                    |                             |   |       |        |
| 4                                                                 | 25 | 0.1   | NaClO <sub>4</sub> | $\lg K: 29.21 (4\text{SF})$ | 0 | MGL   | 1361   |
| 5                                                                 | 25 | 0.5   | NaCl               | $\lg K: 31.0 (3\text{SF})$  | 0 | MGL   | 4358   |
| 6                                                                 | 25 | 0.5   | NaClO <sub>4</sub> | $\lg K: 31.0 (3\text{SF})$  | 0 | MGL   | 4358   |
| 7                                                                 | 25 | 1     | NaBr               | $\lg K: 29.4 (3\text{SF})$  | 0 | MGL   | 4358   |
| 8                                                                 | 25 | 1     | NaCl               | $\lg K: 30.2 (3\text{SF})$  | 3 | MGL   | 4358   |
| 9                                                                 | 25 | 1     | NaClO <sub>4</sub> | $\lg K: 30.2 (3\text{SF})$  | 3 | MGL   | 4358   |

| Reaction No. 17643                                                                                      |    |       |                    |                              |   |       |        |
|---------------------------------------------------------------------------------------------------------|----|-------|--------------------|------------------------------|---|-------|--------|
| $\text{Cu}_2\text{BisTren} + \text{H}_2\text{O} = \text{Cu}_2\text{BisTren}_{\text{OH}-1} + \text{H}^+$ |    |       |                    |                              |   |       |        |
| No                                                                                                      | t  | I-Str | Medium             | Value (Dev)                  | W | Tech. | Ref. # |
| 1                                                                                                       | 25 | 0.1   | NaClO <sub>4</sub> | $\lg K: -10.23 (4\text{SF})$ | 5 | MGL   | 1361   |

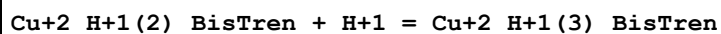
| Reaction No. 17644                                                                                            |    |       |                    |                             |   |       |        |
|---------------------------------------------------------------------------------------------------------------|----|-------|--------------------|-----------------------------|---|-------|--------|
| $\text{Cu}_2(2)\text{BisTren} + \text{H}_2\text{O} = \text{Cu}_2(2)\text{BisTren}_{\text{OH}-1} + \text{H}^+$ |    |       |                    |                             |   |       |        |
| No                                                                                                            | t  | I-Str | Medium             | Value (Dev)                 | W | Tech. | Ref. # |
| 1                                                                                                             | 25 | 0.1   | NaClO <sub>4</sub> | $\lg K: -4.26 (3\text{SF})$ | 6 | MGL   | 1361   |
| 2                                                                                                             | 25 | 0.1   | NaClO <sub>4</sub> | $\lg K: -3.89 (3\text{SF})$ | 5 | MGL   | 4379   |

| Reaction No. 17645                                                             |    |       |                    |                            |   |       |        |
|--------------------------------------------------------------------------------|----|-------|--------------------|----------------------------|---|-------|--------|
| $\text{Cu}_2\text{BisTren} + \text{H}^+ = \text{Cu}_2\text{H}^+\text{BisTren}$ |    |       |                    |                            |   |       |        |
| No                                                                             | t  | I-Str | Medium             | Value (Dev)                | W | Tech. | Ref. # |
| 1                                                                              | 25 | 0     | Inf. Dilution      | $\lg K: 10.0 (3\text{SF})$ | 1 | ELC   |        |
| 2                                                                              | 25 | 0.1   | NaClO <sub>4</sub> | $\lg K: 8.78 (3\text{SF})$ | 0 | MGL   | 1361   |
| 3                                                                              | 25 | 0.1   | NaClO <sub>4</sub> | $\lg K: 7.92 (3\text{SF})$ | 0 | MGL   | 4379   |

| Reaction No. 17646                                                                          |    |       |               |                           |   |       |        |
|---------------------------------------------------------------------------------------------|----|-------|---------------|---------------------------|---|-------|--------|
| $\text{Cu}_2\text{H}^+\text{BisTren} + \text{H}^+ = \text{Cu}_2\text{H}^+(2)\text{BisTren}$ |    |       |               |                           |   |       |        |
| No                                                                                          | t  | I-Str | Medium        | Value (Dev)               | W | Tech. | Ref. # |
| 1                                                                                           | 25 | 0     | Inf. Dilution | $\lg K: 7.9 (2\text{SF})$ | 1 | ELC   |        |

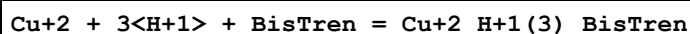
|   |    |     |                    |                |   |     |      |
|---|----|-----|--------------------|----------------|---|-----|------|
| 2 | 25 | 0.1 | NaClO <sub>4</sub> | lgK:7.70 (3SF) | 1 | MGL | 1361 |
| 3 | 25 | 0.1 | NaClO <sub>4</sub> | lgK:7.14 (3SF) | 0 | MGL | 4379 |

## Reaction No. 17647



| No | t  | I-Str | Medium             | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------------------|----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution      | lgK:3.9 (2SF)  | 1 | ELC   |        |
| 2  | 25 | 0.1   | NaClO <sub>4</sub> | lgK:6.87 (3SF) | 0 | MGL   | 1361   |
| 3  | 25 | 0.1   | NaClO <sub>4</sub> | lgK:5.20 (3SF) | 0 | MGL   | 4379   |

## Reaction No. 17648



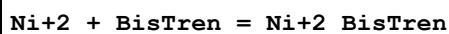
| No | t  | I-Str | Medium             | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------------------|-----------------|---|-------|--------|
| 1  | 25 | 0.025 | NaF                | lgK:43.80 (4SF) | 0 | MGL   | 4358   |
| 2  | 25 | 0.1   | KCl                | lgK:40.09 (4SF) | 4 | MGL   | 1545   |
| 3  | 25 | 0.1   | NaClO <sub>4</sub> | lgK:39.89 (4SF) | 4 | MGL   | 1361   |
| 4  | 25 | 0.1   | NaClO <sub>4</sub> | lgK:38.36 (4SF) | 0 | MGL   | 1545   |

Note (4): Low wgt for internal consistency

|   |    |     |                    |                 |   |     |      |
|---|----|-----|--------------------|-----------------|---|-----|------|
| 5 | 25 | 0.5 | NaCl               | lgK:42.2 (3SF)  | 3 | MGL | 4358 |
| 6 | 25 | 0.5 | NaClO <sub>4</sub> | lgK:42.2 (3SF)  | 3 | MGL | 4358 |
| 7 | 25 | 1   | NaBr               | lgK:42.63 (4SF) | 4 | MGL | 4358 |
| 8 | 25 | 1   | NaCl               | lgK:42.75 (4SF) | 4 | MGL | 4358 |
| 9 | 25 | 1   | NaClO <sub>4</sub> | lgK:41.78 (4SF) | 0 | MGL | 4358 |

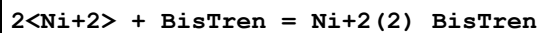
Note (9): Low wgt for internal consistency

## Reaction No. 17649



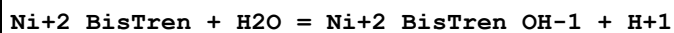
| No | t  | I-Str | Medium             | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------------------|-----------------|---|-------|--------|
| 1  | 25 | 0.1   | NaClO <sub>4</sub> | lgK:11.70 (4SF) | 6 | MGL   | 1361   |

## Reaction No. 17650



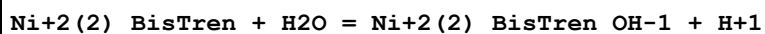
| No | t  | I-Str | Medium             | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------------------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | NaClO <sub>4</sub> | lgK:18.5 (3SF) | 4 | MGL   | 1361   |

## Reaction No. 17651



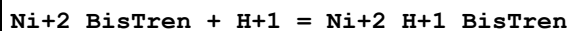
| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.1   | NaClO4 | lgK:-10.3 (3SF) | 6 | MGL   | 1361   |

## Reaction No. 17652



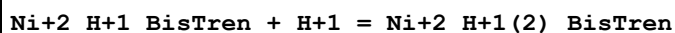
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | NaClO4 | lgK:-6.0 (2SF) | 6 | MGL   | 1361   |

## Reaction No. 17653



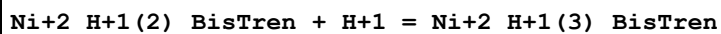
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | NaClO4 | lgK:8.72 (3SF) | 6 | MGL   | 1361   |

## Reaction No. 17654



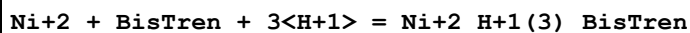
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | NaClO4 | lgK:7.76 (3SF) | 6 | MGL   | 1361   |

## Reaction No. 17655



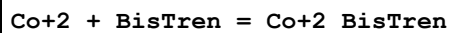
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | NaClO4 | lgK:5.45 (3SF) | 6 | MGL   | 1361   |

## Reaction No. 17656



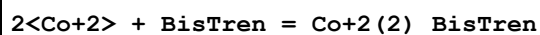
| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.1   | NaClO4 | lgK:33.63 (4SF) | 6 | MGL   | 1361   |

## Reaction No. 17657



| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.1   | NaClO4 | lgK:11.20 (4SF) | 6 | MGL   | 1361   |

## Reaction No. 17658



| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.1   | NaClO4 | lgK:16.80 (4SF) | 6 | MGL   | 1361   |



| Reaction No. 17659                           |    |       |        |                 |   |       |        |
|----------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| Co+2_BisTren + H2O = Co+2_BisTren_OH-1 + H+1 |    |       |        |                 |   |       |        |
| No                                           | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                            | 25 | 0.1   | NaClO4 | lgK:-9.13 (3SF) | 6 | MGL   | 1361   |

| Reaction No. 17660                                 |    |       |        |                 |   |       |        |
|----------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| Co+2(2)_BisTren + H2O = Co+2(2)_BisTren_OH-1 + H+1 |    |       |        |                 |   |       |        |
| No                                                 | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                  | 25 | 0.1   | NaClO4 | lgK:-7.20 (3SF) | 6 | MGL   | 1361   |

| Reaction No. 17661                    |    |       |        |                |   |       |        |
|---------------------------------------|----|-------|--------|----------------|---|-------|--------|
| Co+2_BisTren + H+1 = Co+2_H+1_BisTren |    |       |        |                |   |       |        |
| No                                    | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                     | 25 | 0.1   | NaClO4 | lgK:8.52 (3SF) | 6 | MGL   | 1361   |

| Reaction No. 17662                           |    |       |        |                |   |       |        |
|----------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| Co+2_H+1_BisTren + H+1 = Co+2_H+1(2)_BisTren |    |       |        |                |   |       |        |
| No                                           | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                            | 25 | 0.1   | NaClO4 | lgK:7.16 (3SF) | 6 | MGL   | 1361   |

| Reaction No. 17663                              |    |       |        |                |   |       |        |
|-------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| Co+2_H+1(2)_BisTren + H+1 = Co+2_H+1(3)_BisTren |    |       |        |                |   |       |        |
| No                                              | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                               | 25 | 0.1   | NaClO4 | lgK:6.85 (3SF) | 6 | MGL   | 1361   |

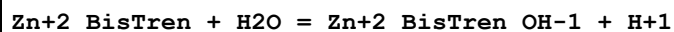
| Reaction No. 17664                            |    |       |        |                 |   |       |        |
|-----------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| Co+2 + BisTren + 3<H+1> = Co+2_H+1(3)_BisTren |    |       |        |                 |   |       |        |
| No                                            | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                             | 25 | 0.1   | NaClO4 | lgK:33.73 (4SF) | 6 | MGL   | 1361   |

| Reaction No. 17665            |    |       |        |                 |   |       |        |
|-------------------------------|----|-------|--------|-----------------|---|-------|--------|
| Zn+2 + BisTren = Zn+2_BisTren |    |       |        |                 |   |       |        |
| No                            | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                             | 25 | 0.1   | NaClO4 | lgK:11.86 (4SF) | 6 | MGL   | 1361   |

| Reaction No. 17666                  |   |       |        |             |   |       |        |
|-------------------------------------|---|-------|--------|-------------|---|-------|--------|
| 2<Zn+2> + BisTren = Zn+2(2)_BisTren |   |       |        |             |   |       |        |
| No                                  | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |

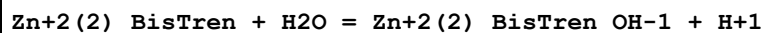
|   |    |     |        |                 |   |     |      |
|---|----|-----|--------|-----------------|---|-----|------|
| 1 | 25 | 0.1 | NaClO4 | lgK:18.22 (4SF) | 6 | MGL | 1361 |
|---|----|-----|--------|-----------------|---|-----|------|

## Reaction No. 17667



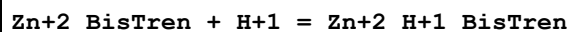
| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.1   | NaClO4 | lgK:-11.1 (3SF) | 6 | MGL   | 1361   |

## Reaction No. 17668



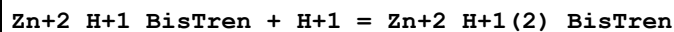
| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.1   | NaClO4 | lgK:-6.00 (3SF) | 6 | MGL   | 1361   |

## Reaction No. 17669



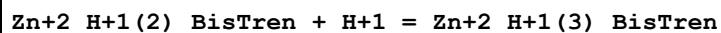
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | NaClO4 | lgK:8.47 (3SF) | 6 | MGL   | 1361   |

## Reaction No. 17670



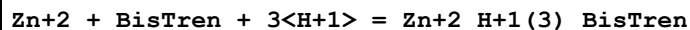
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | NaClO4 | lgK:7.72 (3SF) | 6 | MGL   | 1361   |

## Reaction No. 17671



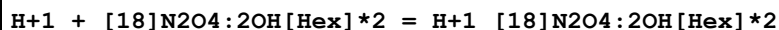
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | NaClO4 | lgK:7.03 (3SF) | 6 | MGL   | 1361   |

## Reaction No. 17672



| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.1   | NaClO4 | lgK:35.08 (4SF) | 6 | MGL   | 1361   |

## Reaction No. 18768



| No | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|--------|-------------------|---|-------|--------|
| 1  | 25 | 0.1   | NaNO3  | lgK:8.88 (0.01SD) | 4 | MGL   | 1711   |

## Reaction No. 18769

| H+1_[18]N2O4:2OH[Hex]*2 + H+1 = H+1(2)_[18]N2O4:2OH[Hex]*2 |    |       |        |                  |   |       |        |
|------------------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| No                                                         | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                          | 25 | 0.1   | NaNO3  | lgK:8.04(0.01SD) | 4 | MGL   | 1711   |

| Reaction No. 18770                                    |    |       |        |                  |   |       |        |
|-------------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| [18]N2O4:2OH[Hex]*2 + Cu+2 = Cu+2_[18]N2O4:2OH[Hex]*2 |    |       |        |                  |   |       |        |
| No                                                    | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                     | 25 | 0.1   | NaNO3  | lgK:7.63(0.02SD) | 6 | MGL   | 1711   |

| Reaction No. 18771                                              |    |       |        |                  |   |       |        |
|-----------------------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| Cu+2_[18]N2O4:2OH[Hex]*2 + OH-1 = Cu+2_[18]N2O4:2OH[Hex]*2_OH-1 |    |       |        |                  |   |       |        |
| No                                                              | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                               | 25 | 0.1   | NaNO3  | lgK:7.24(0.02SD) | 4 | MGL   | 1711   |

| Reaction No. 18772                                    |    |       |        |                  |   |       |        |
|-------------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| [18]N2O4:2OH[Hex]*2 + Zn+2 = Zn+2_[18]N2O4:2OH[Hex]*2 |    |       |        |                  |   |       |        |
| No                                                    | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                     | 25 | 0.1   | NaNO3  | lgK:4.77(0.02SD) | 4 | MGL   | 1711   |

| Reaction No. 18773                                    |    |       |        |                  |   |       |        |
|-------------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| [18]N2O4:2OH[Hex]*2 + Cd+2 = Cd+2_[18]N2O4:2OH[Hex]*2 |    |       |        |                  |   |       |        |
| No                                                    | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                     | 25 | 0.1   | NaNO3  | lgK:8.64(0.02SD) | 6 | MGL   | 1711   |

| Reaction No. 18774                                              |    |       |        |                 |   |       |        |
|-----------------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| Cd+2_[18]N2O4:2OH[Hex]*2 + OH-1 = Cd+2_[18]N2O4:2OH[Hex]*2_OH-1 |    |       |        |                 |   |       |        |
| No                                                              | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                               | 25 | 0.1   | NaNO3  | lgK:5.3(0.07SD) | 4 | MGL   | 1711   |

| Reaction No. 18775                                    |    |       |        |                  |   |       |        |
|-------------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| [18]N2O4:2OH[Hex]*2 + Ca+2 = Ca+2_[18]N2O4:2OH[Hex]*2 |    |       |        |                  |   |       |        |
| No                                                    | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                     | 25 | 0.1   | NaNO3  | lgK:4.72(0.01SD) | 6 | MGL   | 1711   |

| Reaction No. 18776                                    |    |       |        |                  |   |       |        |
|-------------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| [18]N2O4:2OH[Hex]*2 + Sr+2 = Sr+2_[18]N2O4:2OH[Hex]*2 |    |       |        |                  |   |       |        |
| No                                                    | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                     | 25 | 0.1   | NaNO3  | lgK:4.13(0.02SD) | 4 | MGL   | 1711   |

| Reaction No. 18777                                    |    |       |        |                  |   |       |        |
|-------------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| [18]N2O4:2OH[Hex]*2 + Ba+2 = Ba+2_[18]N2O4:2OH[Hex]*2 |    |       |        |                  |   |       |        |
| No                                                    | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                     | 25 | 0.1   | NaNO3  | lgK:4.59(0.01SD) | 6 | MGL   | 1711   |

| Reaction No. 18778                                    |    |       |        |                  |   |       |        |
|-------------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| [18]N2O4:2OH[Hex]*2 + Pb+2 = Pb+2_[18]N2O4:2OH[Hex]*2 |    |       |        |                  |   |       |        |
| No                                                    | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                     | 25 | 0.1   | NaNO3  | lgK:8.59(0.01SD) | 6 | MGL   | 1711   |

| Reaction No. 18779                                              |    |       |        |                 |   |       |        |
|-----------------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| Pb+2_[18]N2O4:2OH[Hex]*2 + OH-1 = Pb+2_[18]N2O4:2OH[Hex]*2_OH-1 |    |       |        |                 |   |       |        |
| No                                                              | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                               | 25 | 0.1   | NaNO3  | lgK:5.2(0.03SD) | 4 | MGL   | 1711   |

| Reaction No. 18780                                    |    |       |        |                 |   |       |        |
|-------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| [18]N2O4:2OH[Hex]*2 + La+3 = La+3_[18]N2O4:2OH[Hex]*2 |    |       |        |                 |   |       |        |
| No                                                    | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                     | 25 | 0.1   | NaNO3  | lgK:4.1(0.08SD) | 4 | MGL   | 1711   |

| Reaction No. 18781                                              |    |       |        |                  |   |       |        |
|-----------------------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| La+3_[18]N2O4:2OH[Hex]*2 + OH-1 = La+3_[18]N2O4:2OH[Hex]*2_OH-1 |    |       |        |                  |   |       |        |
| No                                                              | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                               | 25 | 0.1   | NaNO3  | lgK:5.97(0.02SD) | 4 | MGL   | 1711   |

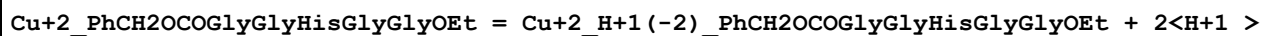
| Reaction No. 18782                                    |    |       |        |                  |   |       |        |
|-------------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| [18]N2O4:2OH[Hex]*2 + Ag+1 = Ag+1_[18]N2O4:2OH[Hex]*2 |    |       |        |                  |   |       |        |
| No                                                    | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                     | 25 | 0.1   | NaNO3  | lgK:5.20(0.01SD) | 6 | MGL   | 1711   |

| Reaction No. 20828                                                |    |       |        |                  |   |       |        |
|-------------------------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| H+1 + PhCH2OCOglyGlyHisGlyGlyOEt = H+1_PhCH2OCOglyGlyHisGlyGlyOEt |    |       |        |                  |   |       |        |
| No                                                                | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                                 | 37 | 0.15  | KNO3   | lgK:6.20(0.01SD) | 6 | MGL   | 1630   |

| Reaction No. 20829                                                  |  |  |  |  |  |  |  |
|---------------------------------------------------------------------|--|--|--|--|--|--|--|
| PhCH2OCOglyGlyHisGlyGlyOEt + Cu+2 = Cu+2_PhCH2OCOglyGlyHisGlyGlyOEt |  |  |  |  |  |  |  |

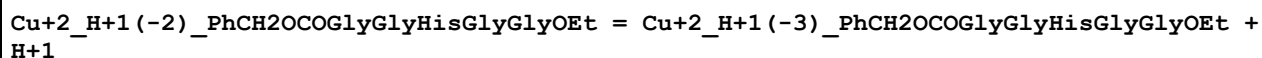
| No | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|--------|-------------------|---|-------|--------|
| 1  | 37 | 0.15  | KNO3   | lgK:3.60 (0.01SD) | 6 | MGL   | 1630   |

## Reaction No. 20830



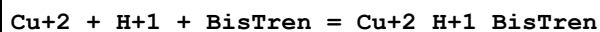
| No | t  | I-Str | Medium | Value (Dev)         | W | Tech. | Ref. # |
|----|----|-------|--------|---------------------|---|-------|--------|
| 1  | 37 | 0.15  | KNO3   | lgK:-11.48 (0.01SD) | 6 | MGL   | 1630   |

## Reaction No. 20831



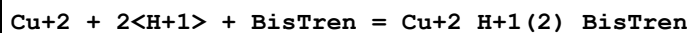
| No | t  | I-Str | Medium | Value (Dev)        | W | Tech. | Ref. # |
|----|----|-------|--------|--------------------|---|-------|--------|
| 1  | 37 | 0.15  | KNO3   | lgK:-7.24 (0.01SD) | 6 | MGL   | 1630   |

## Reaction No. 23945



| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.025 | NaF    | lgK:27.90 (4SF) | 3 | MGL   | 4358   |
| 2  | 25 | 0.1   | KCl    | lgK:24.98 (4SF) | 0 | MGL   | 1545   |
| 3  | 25 | 0.1   | NaClO4 | lgK:25.50 (4SF) | 0 | MGL   | 1545   |
| 4  | 25 | 0.5   | NaCl   | lgK:27.11 (4SF) | 3 | MGL   | 4358   |
| 5  | 25 | 0.5   | NaClO4 | lgK:27.11 (4SF) | 3 | MGL   | 4358   |
| 6  | 25 | 1     | NaBr   | lgK:27.24 (4SF) | 0 | MGL   | 4358   |
| 7  | 25 | 1     | NaCl   | lgK:26.43 (4SF) | 3 | MGL   | 4358   |
| 8  | 25 | 1     | NaClO4 | lgK:27.88 (4SF) | 0 | MGL   | 4358   |

## Reaction No. 23946



| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.025 | NaF    | lgK:36.45 (4SF) | 0 | MGL   | 4358   |
| 2  | 25 | 0.1   | KCl    | lgK:33.10 (4SF) | 0 | MGL   | 1545   |
| 3  | 25 | 0.1   | NaClO4 | lgK:33.01 (4SF) | 0 | MGL   | 1545   |
| 4  | 25 | 0.5   | NaCl   | lgK:35.61 (4SF) | 3 | MGL   | 4358   |
| 5  | 25 | 0.5   | NaClO4 | lgK:35.61 (4SF) | 3 | MGL   | 4358   |
| 6  | 25 | 1     | NaBr   | lgK:35.59 (4SF) | 2 | MGL   | 4358   |
| 7  | 25 | 1     | NaCl   | lgK:35.08 (4SF) | 0 | MGL   | 4358   |
| 8  | 25 | 1     | NaClO4 | lgK:36.03 (4SF) | 2 | MGL   | 4358   |

| Reaction No. 23947                                                                                                      |    |       |        |                |   |       |        |
|-------------------------------------------------------------------------------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| $2\langle\text{Cu}+2\rangle + \text{BisTren} + \text{H}_2\text{O} = \text{Cu}+2(2)_{\text{BisTren\_OH-1}} + \text{H}+1$ |    |       |        |                |   |       |        |
| No                                                                                                                      | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                                                                       | 25 | 0.1   | KCl    | lgK:23.63(4SF) | 0 | MGL   | 1545   |
| 2                                                                                                                       | 25 | 0.1   | NaClO4 | lgK:25.07(4SF) | 0 | MGL   | 1545   |

| Reaction No. 29536                                                              |    |       |        |                   |   |       |        |
|---------------------------------------------------------------------------------|----|-------|--------|-------------------|---|-------|--------|
| $2\langle\text{Pb}+2\rangle + [36]\text{N12} = \text{Pb}+2(2)_{[36]\text{N12}}$ |    |       |        |                   |   |       |        |
| No                                                                              | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                                               | 25 | 0.15  | NaClO4 | lgK:19.97(0.02SD) | 6 | MGL   | 2584   |

| Reaction No. 29537                                                                                     |    |       |        |                   |   |       |        |
|--------------------------------------------------------------------------------------------------------|----|-------|--------|-------------------|---|-------|--------|
| $2\langle\text{Pb}+2\rangle + \text{H}+1 + [36]\text{N12} = \text{Pb}+2(2)_{\text{H}+1}[36]\text{N12}$ |    |       |        |                   |   |       |        |
| No                                                                                                     | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                                                                      | 25 | 0.15  | NaClO4 | lgK:28.36(0.02SD) | 6 | MGL   | 2584   |

| Reaction No. 29538                                                                                                       |    |       |        |                   |   |       |        |
|--------------------------------------------------------------------------------------------------------------------------|----|-------|--------|-------------------|---|-------|--------|
| $2\langle\text{Pb}+2\rangle + 2\langle\text{H}+1\rangle + [36]\text{N12} = \text{Pb}+2(2)_{\text{H}+1(2)}[36]\text{N12}$ |    |       |        |                   |   |       |        |
| No                                                                                                                       | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                                                                                        | 25 | 0.15  | NaClO4 | lgK:35.66(0.02SD) | 6 | MGL   | 2584   |

| Reaction No. 29539                                                                                                       |    |       |        |                  |   |       |        |
|--------------------------------------------------------------------------------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| $2\langle\text{Pb}+2\rangle + 3\langle\text{H}+1\rangle + [36]\text{N12} = \text{Pb}+2(2)_{\text{H}+1(3)}[36]\text{N12}$ |    |       |        |                  |   |       |        |
| No                                                                                                                       | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                                                                                        | 25 | 0.15  | NaClO4 | lgK:41.8(0.03SD) | 6 | MGL   | 2584   |

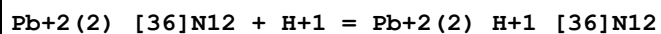
| Reaction No. 29540                                                                                                       |    |       |        |                   |   |       |        |
|--------------------------------------------------------------------------------------------------------------------------|----|-------|--------|-------------------|---|-------|--------|
| $2\langle\text{Pb}+2\rangle + 4\langle\text{H}+1\rangle + [36]\text{N12} = \text{Pb}+2(2)_{\text{H}+1(4)}[36]\text{N12}$ |    |       |        |                   |   |       |        |
| No                                                                                                                       | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                                                                                        | 25 | 0.15  | NaClO4 | lgK:48.23(0.01SD) | 6 | MGL   | 2584   |

| Reaction No. 29541                                                                                                      |    |       |        |                 |   |       |        |
|-------------------------------------------------------------------------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| $2\langle\text{Pb}+2\rangle + [36]\text{N12} + \text{H}_2\text{O} = \text{Pb}+2(2)_{[36]\text{N12\_OH-1}} + \text{H}+1$ |    |       |        |                 |   |       |        |
| No                                                                                                                      | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                                                                       | 25 | 0.15  | NaClO4 | lgK:9.5(0.03SD) | 6 | MGL   | 2584   |

| Reaction No. 29542                                                                                                                                       |  |  |  |  |  |  |  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|--|--|--|--|
| $2\langle\text{Pb}+2\rangle + [36]\text{N12} + 2\langle\text{H}_2\text{O}\rangle = \text{Pb}+2(2)_{[36]\text{N12\_OH-1}(2)} + 2\langle\text{H}+1\rangle$ |  |  |  |  |  |  |  |

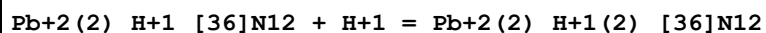
| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 0.15  | NaClO4 | lgK:-1.5(0.03SD) | 6 | MGL   | 2584   |

## Reaction No. 29543



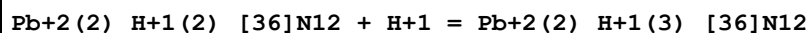
| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.15  | NaClO4 | lgK:8.4(0.04SD) | 6 | MGL   | 2584   |

## Reaction No. 29544



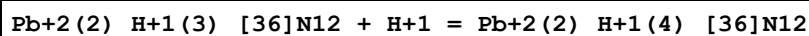
| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.15  | NaClO4 | lgK:7.3(0.04SD) | 6 | MGL   | 2584   |

## Reaction No. 29545



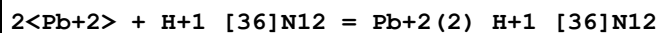
| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.15  | NaClO4 | lgK:6.2(0.05SD) | 6 | MGL   | 2584   |

## Reaction No. 29546



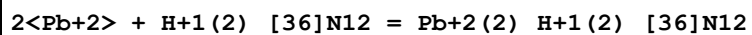
| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.15  | NaClO4 | lgK:6.4(0.04SD) | 6 | MGL   | 2584   |

## Reaction No. 29547



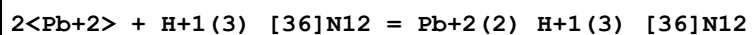
| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 0.15  | NaClO4 | lgK:18.6(0.04SD) | 6 | MGL   | 2584   |

## Reaction No. 29548



| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 0.15  | NaClO4 | lgK:16.3(0.04SD) | 6 | MGL   | 2584   |

## Reaction No. 29549



| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 0.15  | NaClO4 | lgK:13.5(0.05SD) | 6 | MGL   | 2584   |

| Reaction No. 29550                                                                                              |    |       |        |                  |   |       |        |
|-----------------------------------------------------------------------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| $2\langle\text{Pb}+2\rangle + \text{H}+1(4)_{[36]\text{N}12} = \text{Pb}+2(2)_{\text{H}+1(4)_{[36]\text{N}12}}$ |    |       |        |                  |   |       |        |
| No                                                                                                              | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                                                                               | 25 | 0.15  | NaClO4 | lgK:11.0(0.03SD) | 6 | MGL   | 2584   |

| Reaction No. 29551                                                                              |    |       |        |                 |   |       |        |
|-------------------------------------------------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| $\text{Pb}+2(2)_{[36]\text{N}12} + \text{OH}-1 = \text{Pb}+2(2)_{\text{OH}-1_{[36]\text{N}12}}$ |    |       |        |                 |   |       |        |
| No                                                                                              | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                                               | 25 | 0.15  | NaClO4 | lgK:3.2(0.05SD) | 6 | MGL   | 2584   |

| Reaction No. 29552                                                                                               |    |       |        |                 |   |       |        |
|------------------------------------------------------------------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| $\text{Pb}+2(2)_{\text{OH}-1_{[36]\text{N}12}} + \text{OH}-1 = \text{Pb}+2(2)_{\text{OH}-1(2)_{[36]\text{N}12}}$ |    |       |        |                 |   |       |        |
| No                                                                                                               | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                                                                | 25 | 0.15  | NaClO4 | lgK:2.8(0.05SD) | 6 | MGL   | 2584   |

| Reaction No. 29562                                                              |    |       |        |                  |   |       |        |
|---------------------------------------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| $3\langle\text{Pb}+2\rangle + [36]\text{N}12 = \text{Pb}+2(3)_{[36]\text{N}12}$ |    |       |        |                  |   |       |        |
| No                                                                              | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                                               | 25 | 0.15  | NaClO4 | lgK:25.8(0.03SD) | 6 | MGL   | 2584   |

| Reaction No. 29563                                                                                        |    |       |        |                  |   |       |        |
|-----------------------------------------------------------------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| $3\langle\text{Pb}+2\rangle + \text{H}+1 + [36]\text{N}12 = \text{Pb}+2(3)_{\text{H}+1_{[36]\text{N}12}}$ |    |       |        |                  |   |       |        |
| No                                                                                                        | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                                                                         | 25 | 0.15  | NaClO4 | lgK:32.8(0.04SD) | 6 | MGL   | 2584   |

| Reaction No. 29564                                                                                                             |    |       |        |                  |   |       |        |
|--------------------------------------------------------------------------------------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| $3\langle\text{Pb}+2\rangle + [36]\text{N}12 + \text{H}2\text{O} = \text{Pb}+2(3)_{\text{OH}-1_{[36]\text{N}12}} + \text{H}+1$ |    |       |        |                  |   |       |        |
| No                                                                                                                             | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                                                                                              | 25 | 0.15  | NaClO4 | lgK:16.7(0.04SD) | 6 | MGL   | 2584   |

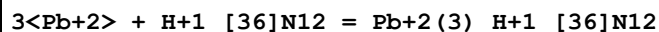
| Reaction No. 29565                                                                                                                                              |    |       |        |                 |   |       |        |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| $3\langle\text{Pb}+2\rangle + [36]\text{N}12 + 2\langle\text{H}2\text{O}\rangle = \text{Pb}+2(3)_{\text{OH}-1(2)_{[36]\text{N}12}} + 2\langle\text{H}+1\rangle$ |    |       |        |                 |   |       |        |
| No                                                                                                                                                              | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                                                                                                               | 25 | 0.15  | NaClO4 | lgK:7.0(0.04SD) | 6 | MGL   | 2584   |

| Reaction No. 29566                                                                                                                                              |   |       |        |             |   |       |        |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|---|-------|--------|-------------|---|-------|--------|
| $3\langle\text{Pb}+2\rangle + [36]\text{N}12 + 3\langle\text{H}2\text{O}\rangle = \text{Pb}+2(3)_{\text{OH}-1(3)_{[36]\text{N}12}} + 3\langle\text{H}+1\rangle$ |   |       |        |             |   |       |        |
| No                                                                                                                                                              | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |



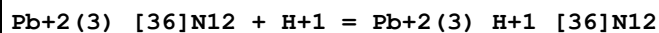
|   |    |      |        |                  |   |     |      |
|---|----|------|--------|------------------|---|-----|------|
| 1 | 25 | 0.15 | NaClO4 | lgK:-3.7(0.07SD) | 6 | MGL | 2584 |
|---|----|------|--------|------------------|---|-----|------|

## Reaction No. 29567



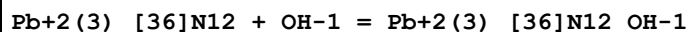
| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 0.15  | NaClO4 | lgK:23.0(0.06SD) | 6 | MGL   | 2584   |

## Reaction No. 29568



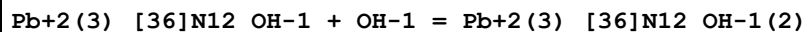
| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.15  | NaClO4 | lgK:7.0(0.07SD) | 6 | MGL   | 2584   |

## Reaction No. 29569



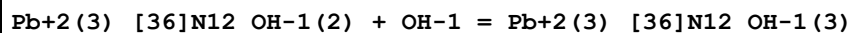
| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.15  | NaClO4 | lgK:4.7(0.06SD) | 6 | MGL   | 2584   |

## Reaction No. 29570



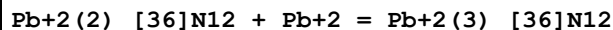
| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.15  | NaClO4 | lgK:4.0(0.06SD) | 6 | MGL   | 2584   |

## Reaction No. 29571



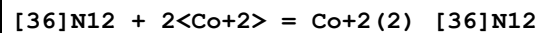
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.15  | NaClO4 | lgK:3.1(0.1SD) | 6 | MGL   | 2584   |

## Reaction No. 29572



| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.15  | NaClO4 | lgK:5.8(0.05SD) | 6 | MGL   | 2584   |

## Reaction No. 31216



| No | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|--------|-------------------|---|-------|--------|
| 1  | 25 | 0.15  | NaClO4 | lgK:24.55(0.02SD) | 6 | MGL   | 2726   |

## Reaction No. 31217

| 4<H+1> + [36]N12 + 2<Co+2> = Co+2(2)_H+1(4)_[36]N12 |    |       |        |                   |   |       |        |
|-----------------------------------------------------|----|-------|--------|-------------------|---|-------|--------|
| No                                                  | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                   | 25 | 0.15  | NaClO4 | lgK:48.76(0.02SD) | 6 | MGL   | 2726   |

| Reaction No. 31218                                  |    |       |        |                  |   |       |        |
|-----------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| 3<H+1> + [36]N12 + 2<Co+2> = Co+2(2)_H+1(3)_[36]N12 |    |       |        |                  |   |       |        |
| No                                                  | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                   | 25 | 0.15  | NaClO4 | lgK:43.4(0.04SD) | 6 | MGL   | 2726   |

| Reaction No. 31219                                  |    |       |        |                  |   |       |        |
|-----------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| 2<H+1> + [36]N12 + 2<Co+2> = Co+2(2)_H+1(2)_[36]N12 |    |       |        |                  |   |       |        |
| No                                                  | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                   | 25 | 0.15  | NaClO4 | lgK:37.6(0.04SD) | 6 | MGL   | 2726   |

| Reaction No. 31220                            |    |       |        |                  |   |       |        |
|-----------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| H+1 + [36]N12 + 2<Co+2> = Co+2(2)_H+1_[36]N12 |    |       |        |                  |   |       |        |
| No                                            | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                             | 25 | 0.15  | NaClO4 | lgK:31.3(0.03SD) | 6 | MGL   | 2726   |

| Reaction No. 31221                                   |    |       |        |                  |   |       |        |
|------------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| 2<Co+2> + [36]N12 + H2O = Co+2(2)_[36]N12_OH-1 + H+1 |    |       |        |                  |   |       |        |
| No                                                   | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                    | 25 | 0.15  | NaClO4 | lgK:13.9(0.03SD) | 6 | MGL   | 2726   |

| Reaction No. 33212                                                |    |       |        |                  |   |       |        |
|-------------------------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| 2<H+1> + 2<AspArgValTyr-3> + Ni+2 = Ni+2_H+1(2)_AspArgValTyr-3(2) |    |       |        |                  |   |       |        |
| No                                                                | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                                 | 25 | 0.1   | KNO3   | lgK:29.1(0.04SD) | 5 | MGL   | 2974   |

| Reaction No. 33409                                    |    |       |        |                 |   |       |        |
|-------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| H+1 + AspArgValTyr-3 + Ni+2 = Ni+2_H+1_AspArgValTyr-3 |    |       |        |                 |   |       |        |
| No                                                    | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                     | 25 | 0.1   | KNO3   | lgK:14.5(0.1SD) | 5 | MGL   | 2974   |

| Reaction No. 33410                                          |    |       |        |                  |   |       |        |
|-------------------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| H+1 + 2<AspArgValTyr-3> + Ni+2 = Ni+2_H+1_AspArgValTyr-3(2) |    |       |        |                  |   |       |        |
| No                                                          | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                           | 25 | 0.1   | KNO3   | lgK:20.0(0.08SD) | 5 | MGL   | 2974   |

| Reaction No. 33412                                           |    |       |        |                   |   |       |        |
|--------------------------------------------------------------|----|-------|--------|-------------------|---|-------|--------|
| AspArgValTyr-3 + Ni+2 = 2<H+1> + Ni+2_H+1(-2)_AspArgValTyr-3 |    |       |        |                   |   |       |        |
| No                                                           | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                            | 25 | 0.1   | KNO3   | lgK:-12.2(0.08SD) | 5 | MGL   | 2974   |

| Reaction No. 33768                          |    |       |        |                  |   |       |        |
|---------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| H+1 + 2*(ProProGly)-1 = H+1_2*(ProProGly)-1 |    |       |        |                  |   |       |        |
| No                                          | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                           | 25 | 0.1   | KNO3   | lgK:9.13(0.01SD) | 7 | MGL   | 1177   |

| Reaction No. 33769                                |    |       |        |                   |   |       |        |
|---------------------------------------------------|----|-------|--------|-------------------|---|-------|--------|
| 2<H+1> + 2*(ProProGly)-1 = H+1(2)_2*(ProProGly)-1 |    |       |        |                   |   |       |        |
| No                                                | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                 | 25 | 0.1   | KNO3   | lgK:12.57(0.02SD) | 7 | MGL   | 1177   |

| Reaction No. 33770                            |    |       |        |                  |   |       |        |
|-----------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| 2*(ProProGly)-1 + Cu+2 = Cu+2_2*(ProProGly)-1 |    |       |        |                  |   |       |        |
| No                                            | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                             | 25 | 0.1   | KNO3   | lgK:6.27(0.01SD) | 7 | MGL   | 1177   |

| Reaction No. 33771                                          |    |       |        |                  |   |       |        |
|-------------------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| 2*(ProProGly)-1 + Cu+2 = H+1 + Cu+2_H+1(-1)_2*(ProProGly)-1 |    |       |        |                  |   |       |        |
| No                                                          | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                           | 25 | 0.1   | KNO3   | lgK:-1.4(0.03SD) | 7 | MGL   | 1177   |

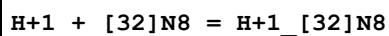
| Reaction No. 33772                                             |    |       |        |                    |   |       |        |
|----------------------------------------------------------------|----|-------|--------|--------------------|---|-------|--------|
| 2*(ProProGly)-1 + Cu+2 = 2<H+1> + Cu+2_H+1(-2)_2*(ProProGly)-1 |    |       |        |                    |   |       |        |
| No                                                             | t  | I-Str | Medium | Value (Dev)        | W | Tech. | Ref. # |
| 1                                                              | 25 | 0.1   | KNO3   | lgK:-11.49(0.01SD) | 7 | MGL   | 1177   |

| Reaction No. 33773                                  |    |       |        |                   |   |       |        |
|-----------------------------------------------------|----|-------|--------|-------------------|---|-------|--------|
| 2<2*(ProProGly)-1> + Cu+2 = Cu+2_2*(ProProGly)-1(2) |    |       |        |                   |   |       |        |
| No                                                  | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                   | 25 | 0.1   | KNO3   | lgK:11.19(0.02SD) | 7 | MGL   | 1177   |

| Reaction No. 33829                                            |  |  |  |  |  |  |  |
|---------------------------------------------------------------|--|--|--|--|--|--|--|
| Co+2(2)_BisTren + O2(g) + H2O = H+1 + Co+2(2)_BisTren_O2_OH-1 |  |  |  |  |  |  |  |

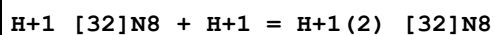
| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.1   | NaClO4 | lgK:-6.00 (3SF) | 5 | MGL   | 2908   |

## Reaction No. 35739



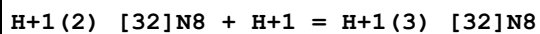
| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 22 | 0.1   | Me4NC1 | lgK:10.70 (4SF) | 5 | MGL   | 4382   |
| 2  | 25 | 0.1   | NaTs   | lgK:10.65 (4SF) | 5 | MPT   | 3280   |

## Reaction No. 35740



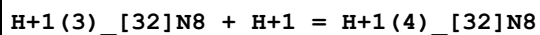
| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 22 | 0.1   | Me4NC1 | lgK:10.45 (4SF) | 5 | MGL   | 4382   |
| 2  | 25 | 0.1   | NaTs   | lgK:10.55 (4SF) | 5 | MPT   | 3280   |

## Reaction No. 35741



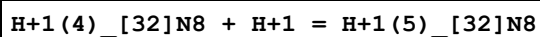
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 22 | 0.1   | Me4NC1 | lgK:9.65 (3SF) | 5 | MGL   | 4382   |
| 2  | 25 | 0.1   | NaTs   | lgK:9.70 (3SF) | 5 | MPT   | 3280   |

## Reaction No. 35742



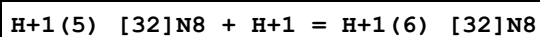
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 22 | 0.1   | Me4NC1 | lgK:9.00 (3SF) | 5 | MGL   | 4382   |
| 2  | 25 | 0.1   | NaTs   | lgK:9.20 (3SF) | 5 | MPT   | 3280   |

## Reaction No. 35743



| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 22 | 0.1   | Me4NC1 | lgK:8.05 (3SF) | 5 | MGL   | 4382   |
| 2  | 25 | 0.1   | NaTs   | lgK:8.20 (3SF) | 5 | MPT   | 3280   |

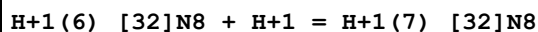
## Reaction No. 35744



| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 22 | 0.1   | Me4NC1 | lgK:7.50 (3SF) | 5 | MGL   | 4382   |

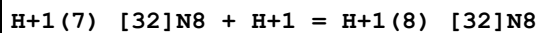
|   |    |     |      |                |   |     |      |
|---|----|-----|------|----------------|---|-----|------|
| 2 | 25 | 0.1 | NaTs | lgK:7.55 (3SF) | 5 | MPT | 3280 |
|---|----|-----|------|----------------|---|-----|------|

## Reaction No. 35745



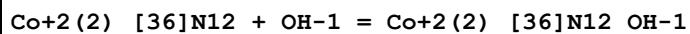
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 22 | 0.1   | Me4NC1 | lgK:6.95 (3SF) | 5 | MGL   | 4382   |
| 2  | 25 | 0.1   | NaTs   | lgK:6.85 (3SF) | 5 | MPT   | 3280   |

## Reaction No. 35746



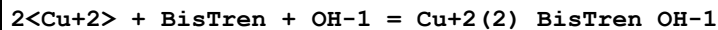
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 22 | 0.1   | Me4NC1 | lgK:6.45 (3SF) | 5 | MGL   | 4382   |
| 2  | 25 | 0.1   | NaTs   | lgK:6.50 (3SF) | 5 | MPT   | 3280   |

## Reaction No. 36098



| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.15  | NaClO4 | lgK:3.05 (3SF) | 5 | MGL   | 2726   |

## Reaction No. 38919

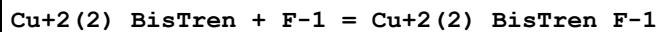


| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.025 | NaF    | lgK:40.33 (4SF) | 5 | MGL   | 4358   |
| 2  | 25 | 0.5   | NaCl   | lgK:40.33 (4SF) | 5 | MGL   | 4358   |
| 3  | 25 | 0.5   | NaClO4 | lgK:40.33 (4SF) | 5 | MGL   | 4358   |
| 4  | 25 | 1     | NaBr   | lgK:40.47 (4SF) | 5 | MGL   | 4358   |
| 5  | 25 | 1     | NaCl   | lgK:39.36 (4SF) | 0 | MGL   | 4358   |

Note (5): Low wgt for internal consistency

|   |    |   |        |                 |   |     |      |
|---|----|---|--------|-----------------|---|-----|------|
| 6 | 25 | 1 | NaClO4 | lgK:40.76 (4SF) | 5 | MGL | 4358 |
|---|----|---|--------|-----------------|---|-----|------|

## Reaction No. 38920



| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | NaClO4 | lgK:4.5 (2SF)  | 5 | MGL   | 4379   |
| 2  | 25 | 1     | NaClO4 | lgK:4.71 (3SF) | 5 | MGL   | 4358   |

## Reaction No. 38921

| Cu+2(2)_BisTren + Cl-1 = Cu+2(2)_BisTren_Cl-1 |    |       |        |                |   |       |        |
|-----------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| No                                            | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                             | 25 | 1     | NaClO4 | lgK:2.11 (3SF) | 5 | MGL   | 4358   |

| Reaction No. 38922                          |    |       |        |               |   |       |        |
|---------------------------------------------|----|-------|--------|---------------|---|-------|--------|
| Cu+2(2)_BisTren + I-1 = Cu+2(2)_BisTren_I-1 |    |       |        |               |   |       |        |
| No                                          | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                           | 25 | 1     | NaClO4 | lgK:3.2 (2SF) | 5 | MGL   | 4358   |

| Reaction No. 38923                                  |    |       |        |                |   |       |        |
|-----------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| Cu+2_H+1(3)_BisTren + F-1 = Cu+2_H+1(3)_BisTren_F-1 |    |       |        |                |   |       |        |
| No                                                  | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                   | 25 | 0.1   | NaClO4 | lgK:4.1 (2SF)  | 5 | MGL   | 4379   |
| 2                                                   | 25 | 1     | NaClO4 | lgK:3.20 (3SF) | 5 | MGL   | 4358   |

| Reaction No. 38924                                    |    |       |        |                |   |       |        |
|-------------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| Cu+2_H+1(3)_BisTren + Cl-1 = Cu+2_H+1(3)_BisTren_Cl-1 |    |       |        |                |   |       |        |
| No                                                    | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                     | 25 | 1     | NaClO4 | lgK:2.27 (3SF) | 5 | MGL   | 4358   |

| Reaction No. 38925                                    |    |       |        |                |   |       |        |
|-------------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| Cu+2_H+1(3)_BisTren + Br-1 = Cu+2_H+1(3)_BisTren_Br-1 |    |       |        |                |   |       |        |
| No                                                    | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                     | 25 | 1     | NaClO4 | lgK:1.72 (3SF) | 5 | MGL   | 4358   |

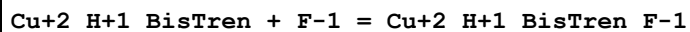
| Reaction No. 38926                                  |    |       |        |                |   |       |        |
|-----------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| Cu+2_H+1(2)_BisTren + F-1 = Cu+2_H+1(2)_BisTren_F-1 |    |       |        |                |   |       |        |
| No                                                  | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                   | 25 | 0.1   | NaClO4 | lgK:3.0 (2SF)  | 5 | MGL   | 4379   |
| 2                                                   | 25 | 1     | NaClO4 | lgK:2.00 (3SF) | 5 | MGL   | 4358   |

| Reaction No. 38927                                    |    |       |        |                |   |       |        |
|-------------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| Cu+2_H+1(2)_BisTren + Cl-1 = Cu+2_H+1(2)_BisTren_Cl-1 |    |       |        |                |   |       |        |
| No                                                    | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                     | 25 | 1     | NaClO4 | lgK:0.54 (2SF) | 5 | MGL   | 4358   |

| Reaction No. 38928                                    |  |  |  |  |  |  |  |
|-------------------------------------------------------|--|--|--|--|--|--|--|
| Cu+2_H+1(2)_BisTren + Br-1 = Cu+2_H+1(2)_BisTren_Br-1 |  |  |  |  |  |  |  |

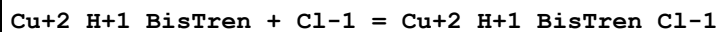
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 1     | NaClO4 | lgK:-0.2 (1SF) | 5 | MGL   | 4358   |

## Reaction No. 38929



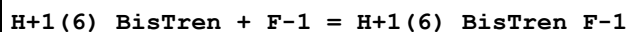
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | NaClO4 | lgK:2.7 (2SF)  | 5 | MGL   | 4379   |
| 2  | 25 | 1     | NaClO4 | lgK:1.78 (3SF) | 5 | MGL   | 4358   |

## Reaction No. 38930



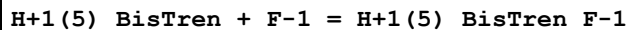
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 1     | NaClO4 | lgK:-0.7 (1SF) | 5 | MGL   | 4358   |

## Reaction No. 38943



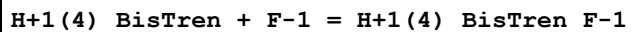
| No | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|--------|---------------|---|-------|--------|
| 1  | 25 | 0.1   | NaClO4 | lgK:3.9 (2SF) | 5 | MGL   | 4379   |

## Reaction No. 38944



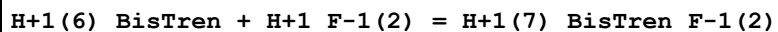
| No | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|--------|---------------|---|-------|--------|
| 1  | 25 | 0.1   | NaClO4 | lgK:3.6 (2SF) | 5 | MGL   | 4379   |

## Reaction No. 38945



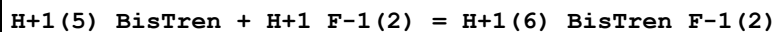
| No | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|--------|---------------|---|-------|--------|
| 1  | 25 | 0.1   | NaClO4 | lgK:2.0 (2SF) | 5 | MGL   | 4379   |

## Reaction No. 38946



| No | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|--------|---------------|---|-------|--------|
| 1  | 25 | 0.1   | NaClO4 | lgK:6.4 (2SF) | 5 | MGL   | 4379   |

## Reaction No. 38947



| No | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|--------|---------------|---|-------|--------|
| 1  | 25 | 0.1   | NaClO4 | lgK:5.3 (2SF) | 5 | MGL   | 4379   |

| Reaction No. 38950                    |    |       |               |                |   |       |        |
|---------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| Cu+2_BisTren + Cu+2 = Cu+2(2)_BisTren |    |       |               |                |   |       |        |
| No                                    | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                     | 25 | 0     | Inf. Dilution | lgK:12.2(3SF)  | 1 | ELC   |        |
| 2                                     | 25 | 0.1   | NaClO4        | lgK:10.73(4SF) | 0 | MGL   | 4379   |

| Reaction No. 38962                            |    |       |        |              |   |       |        |
|-----------------------------------------------|----|-------|--------|--------------|---|-------|--------|
| SO4-2 + [32]N8 + 8<H+1> = H+1(8)_[32]N8_SO4-2 |    |       |        |              |   |       |        |
| No                                            | t  | I-Str | Medium | Value (Dev)  | W | Tech. | Ref. # |
| 1                                             | 23 | 0.1   | Me4NC1 | lgK:4.0(2SF) | 5 | MGL   | 4382   |

| Reaction No. 38963                                  |    |       |               |              |   |       |        |
|-----------------------------------------------------|----|-------|---------------|--------------|---|-------|--------|
| Oxalic-2 + [32]N8 + 8<H+1> = H+1(8)_[32]N8_Oxalic-2 |    |       |               |              |   |       |        |
| No                                                  | t  | I-Str | Medium        | Value (Dev)  | W | Tech. | Ref. # |
| 1                                                   | 23 | 0.1   | Me4NC1        | lgK:3.7(2SF) | 5 | MGL   | 4382   |
| 2                                                   | 25 | 0     | Inf. Dilution | lgK:3.8(2SF) | 1 | ESC   |        |

| Reaction No. 38964                                    |    |       |        |              |   |       |        |
|-------------------------------------------------------|----|-------|--------|--------------|---|-------|--------|
| Malonic-2 + [32]N8 + 8<H+1> = H+1(8)_[32]N8_Malonic-2 |    |       |        |              |   |       |        |
| No                                                    | t  | I-Str | Medium | Value (Dev)  | W | Tech. | Ref. # |
| 1                                                     | 22 | 0.1   | Me4NC1 | lgK:3.9(2SF) | 5 | MGL   | 4382   |

| Reaction No. 38965                                      |    |       |        |              |   |       |        |
|---------------------------------------------------------|----|-------|--------|--------------|---|-------|--------|
| Succinic-2 + [32]N8 + 8<H+1> = H+1(8)_[32]N8_Succinic-2 |    |       |        |              |   |       |        |
| No                                                      | t  | I-Str | Medium | Value (Dev)  | W | Tech. | Ref. # |
| 1                                                       | 23 | 0.1   | Me4NC1 | lgK:3.6(2SF) | 5 | MGL   | 4382   |

| Reaction No. 38966                                  |    |       |               |              |   |       |        |
|-----------------------------------------------------|----|-------|---------------|--------------|---|-------|--------|
| Maleic-2 + [32]N8 + 8<H+1> = H+1(8)_[32]N8_Maleic-2 |    |       |               |              |   |       |        |
| No                                                  | t  | I-Str | Medium        | Value (Dev)  | W | Tech. | Ref. # |
| 1                                                   | 23 | 0.1   | Me4NC1        | lgK:4.1(2SF) | 5 | MGL   | 4382   |
| 2                                                   | 25 | 0     | Inf. Dilution | lgK:4.3(2SF) | 1 | ESC   |        |

| Reaction No. 38967                                    |    |       |        |              |   |       |        |
|-------------------------------------------------------|----|-------|--------|--------------|---|-------|--------|
| Fumaric-2 + [32]N8 + 8<H+1> = H+1(8)_[32]N8_Fumaric-2 |    |       |        |              |   |       |        |
| No                                                    | t  | I-Str | Medium | Value (Dev)  | W | Tech. | Ref. # |
| 1                                                     | 23 | 0.1   | Me4NC1 | lgK:2.9(2SF) | 5 | MGL   | 4382   |



| Reaction No. 38968                                     |    |       |        |              |   |       |        |
|--------------------------------------------------------|----|-------|--------|--------------|---|-------|--------|
| Squaric-2 + [32]N8 + 8<H+1> = H+1(8)_ [32]N8_Squaric-2 |    |       |        |              |   |       |        |
| No                                                     | t  | I-Str | Medium | Value (Dev)  | W | Tech. | Ref. # |
| 1                                                      | 23 | 0.1   | Me4NC1 | lgK:3.6(2SF) | 5 | MGL   | 4382   |

| Reaction No. 38969                                   |    |       |        |              |   |       |        |
|------------------------------------------------------|----|-------|--------|--------------|---|-------|--------|
| Citric-3 + [32]N8 + 8<H+1> = H+1(8)_ [32]N8_Citric-3 |    |       |        |              |   |       |        |
| No                                                   | t  | I-Str | Medium | Value (Dev)  | W | Tech. | Ref. # |
| 1                                                    | 23 | 0.1   | Me4NC1 | lgK:7.6(2SF) | 5 | MGL   | 4382   |

| Reaction No. 38970                                                 |    |       |        |              |   |       |        |
|--------------------------------------------------------------------|----|-------|--------|--------------|---|-------|--------|
| 135BenzTriCOO-3 + [32]N8 + 8<H+1> = H+1(8)_ [32]N8_135BenzTriCOO-3 |    |       |        |              |   |       |        |
| No                                                                 | t  | I-Str | Medium | Value (Dev)  | W | Tech. | Ref. # |
| 1                                                                  | 23 | 0.1   | Me4NC1 | lgK:6.1(2SF) | 5 | MGL   | 4382   |

| Reaction No. 38971                               |    |       |        |              |   |       |        |
|--------------------------------------------------|----|-------|--------|--------------|---|-------|--------|
| 8<H+1> + [32]N8 + 5AMP-2 = H+1(8)_ [32]N8_5AMP-2 |    |       |        |              |   |       |        |
| No                                               | t  | I-Str | Medium | Value (Dev)  | W | Tech. | Ref. # |
| 1                                                | 25 | 0.1   | Me4NC1 | lgK:4.1(2SF) | 5 | MGL   | 4382   |

| Reaction No. 38972                               |    |       |        |              |   |       |        |
|--------------------------------------------------|----|-------|--------|--------------|---|-------|--------|
| 5ADP-3 + [32]N8 + 8<H+1> = H+1(8)_ [32]N8_5ADP-3 |    |       |        |              |   |       |        |
| No                                               | t  | I-Str | Medium | Value (Dev)  | W | Tech. | Ref. # |
| 1                                                | 23 | 0.1   | Me4NC1 | lgK:7.5(2SF) | 5 | MGL   | 4382   |

| Reaction No. 38973                               |    |       |               |              |   |       |        |
|--------------------------------------------------|----|-------|---------------|--------------|---|-------|--------|
| 5ATP-4 + [32]N8 + 8<H+1> = H+1(8)_ [32]N8_5ATP-4 |    |       |               |              |   |       |        |
| No                                               | t  | I-Str | Medium        | Value (Dev)  | W | Tech. | Ref. # |
| 1                                                | 23 | 0.1   | Me4NC1        | lgK:8.5(2SF) | 5 | MGL   | 4382   |
| 2                                                | 25 | 0     | Inf. Dilution | lgK:8.7(2SF) | 1 | ESC   |        |

| Reaction No. 39043                   |    |       |                                  |                   |   |       |        |
|--------------------------------------|----|-------|----------------------------------|-------------------|---|-------|--------|
| [36]N12 + 2<Zn+2> = Zn+2(2)_ [36]N12 |    |       |                                  |                   |   |       |        |
| No                                   | t  | I-Str | Medium                           | Value (Dev)       | W | Tech. | Ref. # |
| 1                                    | 25 | 0.15  | NaClO4                           | lgK:26.27(4SF)    | 6 | MPT   | 4257   |
| 2                                    | 25 | 0.15  | DMSO:Water<br>80:20Vol<br>NaClO4 | lgK:32.25(0.01SD) | 5 | MGL   | 2548   |

| Reaction No. 39044                               |    |       |                                  |                   |   |       |        |
|--------------------------------------------------|----|-------|----------------------------------|-------------------|---|-------|--------|
| $H+1 + [36]N12 + 2<Zn+2> = Zn+2(2)_{H+1}[36]N12$ |    |       |                                  |                   |   |       |        |
| No                                               | t  | I-Str | Medium                           | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                | 25 | 0.15  | NaClO4                           | lgK:32.83(4SF)    | 6 | MPT   | 4257   |
| 2                                                | 25 | 0.15  | DMSO:Water<br>80:20Vol<br>NaClO4 | lgK:37.77(0.01SD) | 5 | MGL   | 2548   |

| Reaction No. 39045                                     |    |       |                                  |                   |   |       |        |
|--------------------------------------------------------|----|-------|----------------------------------|-------------------|---|-------|--------|
| $2<H+1> + [36]N12 + 2<Zn+2> = Zn+2(2)_{H+1(2)}[36]N12$ |    |       |                                  |                   |   |       |        |
| No                                                     | t  | I-Str | Medium                           | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                      | 25 | 0.15  | NaClO4                           | lgK:39.16(4SF)    | 6 | MPT   | 4257   |
| 2                                                      | 25 | 0.15  | DMSO:Water<br>80:20Vol<br>NaClO4 | lgK:42.67(0.02SD) | 5 | MGL   | 2548   |

| Reaction No. 39046                                     |    |       |                                  |                  |   |       |        |
|--------------------------------------------------------|----|-------|----------------------------------|------------------|---|-------|--------|
| $3<H+1> + [36]N12 + 2<Zn+2> = Zn+2(2)_{H+1(3)}[36]N12$ |    |       |                                  |                  |   |       |        |
| No                                                     | t  | I-Str | Medium                           | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                      | 25 | 0.15  | NaClO4                           | lgK:44.81(4SF)   | 6 | MPT   | 4257   |
| 2                                                      | 25 | 0.15  | DMSO:Water<br>80:20Vol<br>NaClO4 | lgK:47.7(0.03SD) | 5 | MGL   | 2548   |

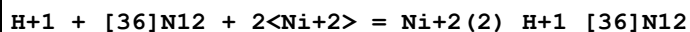
| Reaction No. 39047                                     |    |       |                                  |                  |   |       |        |
|--------------------------------------------------------|----|-------|----------------------------------|------------------|---|-------|--------|
| $4<H+1> + [36]N12 + 2<Zn+2> = Zn+2(2)_{H+1(4)}[36]N12$ |    |       |                                  |                  |   |       |        |
| No                                                     | t  | I-Str | Medium                           | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                      | 25 | 0.15  | NaClO4                           | lgK:49.99(4SF)   | 6 | MPT   | 4257   |
| 2                                                      | 25 | 0.15  | DMSO:Water<br>80:20Vol<br>NaClO4 | lgK:51.8(0.05SD) | 5 | MGL   | 2548   |

| Reaction No. 39048                                         |    |       |        |                |   |       |        |
|------------------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| $2<Zn+2> + [36]N12 + H2O = Zn+2(2)_{[36]N12_{OH-1}} + H+1$ |    |       |        |                |   |       |        |
| No                                                         | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                          | 25 | 0.15  | NaClO4 | lgK:16.09(4SF) | 6 | MPT   | 4257   |

| Reaction No. 39065                      |   |       |        |             |   |       |        |
|-----------------------------------------|---|-------|--------|-------------|---|-------|--------|
| $[36]N12 + 2<Ni+2> = Ni+2(2)_{[36]N12}$ |   |       |        |             |   |       |        |
| No                                      | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |

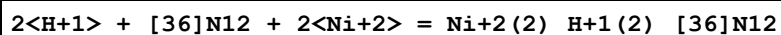
|   |    |      |        |                  |   |     |      |
|---|----|------|--------|------------------|---|-----|------|
| 1 | 25 | 0.15 | NaClO4 | lgK:32.1(0.03SD) | 6 | MPT | 4258 |
|---|----|------|--------|------------------|---|-----|------|

## Reaction No. 39066



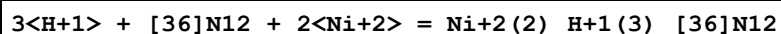
| No | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|--------|-------------------|---|-------|--------|
| 1  | 25 | 0.15  | NaClO4 | lgK:38.62(0.02SD) | 6 | MPT   | 4258   |

## Reaction No. 39067



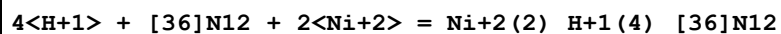
| No | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|--------|-------------------|---|-------|--------|
| 1  | 25 | 0.15  | NaClO4 | lgK:44.46(0.01SD) | 6 | MPT   | 4258   |

## Reaction No. 39068



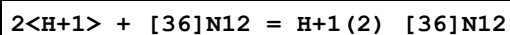
| No | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|--------|-------------------|---|-------|--------|
| 1  | 25 | 0.15  | NaClO4 | lgK:48.95(0.02SD) | 6 | MPT   | 4258   |

## Reaction No. 39069



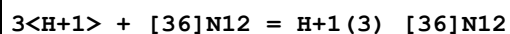
| No | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|--------|-------------------|---|-------|--------|
| 1  | 25 | 0.15  | NaClO4 | lgK:53.42(0.02SD) | 6 | MPT   | 4258   |

## Reaction No. 39323



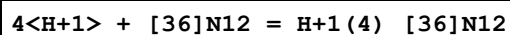
| No | t  | I-Str | Medium                           | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|----------------------------------|-------------------|---|-------|--------|
| 1  | 25 | 0.15  | DMSO:Water<br>80:20Vol<br>NaClO4 | lgK:19.87(0.01SD) | 5 | MGL   | 2548   |

## Reaction No. 39324



| No | t  | I-Str | Medium                           | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|----------------------------------|-------------------|---|-------|--------|
| 1  | 25 | 0.15  | DMSO:Water<br>80:20Vol<br>NaClO4 | lgK:28.89(0.02SD) | 5 | MGL   | 2548   |

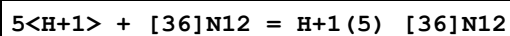
## Reaction No. 39325



| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|----|---|-------|--------|-------------|---|-------|--------|
|----|---|-------|--------|-------------|---|-------|--------|

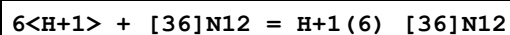
|   |    |      |                                  |                   |   |     |      |
|---|----|------|----------------------------------|-------------------|---|-----|------|
| 1 | 25 | 0.15 | DMSO:Water<br>80:20Vol<br>NaClO4 | lgK:37.36(0.01SD) | 5 | MGL | 2548 |
|---|----|------|----------------------------------|-------------------|---|-----|------|

## Reaction No. 39326



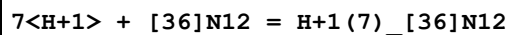
| No | t  | I-Str | Medium                           | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|----------------------------------|-------------------|---|-------|--------|
| 1  | 25 | 0.15  | DMSO:Water<br>80:20Vol<br>NaClO4 | lgK:45.17(0.02SD) | 5 | MGL   | 2548   |

## Reaction No. 39327



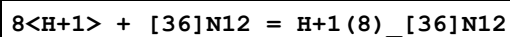
| No | t  | I-Str | Medium                           | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|----------------------------------|-------------------|---|-------|--------|
| 1  | 25 | 0.15  | DMSO:Water<br>80:20Vol<br>NaClO4 | lgK:52.34(0.02SD) | 5 | MGL   | 2548   |

## Reaction No. 39328



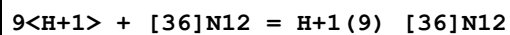
| No | t  | I-Str | Medium                           | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|----------------------------------|-------------------|---|-------|--------|
| 1  | 25 | 0.15  | DMSO:Water<br>80:20Vol<br>NaClO4 | lgK:57.67(0.02SD) | 5 | MGL   | 2548   |

## Reaction No. 39329



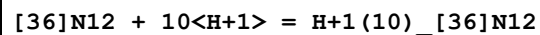
| No | t  | I-Str | Medium                           | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|----------------------------------|-------------------|---|-------|--------|
| 1  | 25 | 0.15  | DMSO:Water<br>80:20Vol<br>NaClO4 | lgK:61.48(0.02SD) | 5 | MGL   | 2548   |

## Reaction No. 39330



| No | t  | I-Str | Medium                           | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|----------------------------------|------------------|---|-------|--------|
| 1  | 25 | 0.15  | DMSO:Water<br>80:20Vol<br>NaClO4 | lgK:64.3(0.03SD) | 5 | MGL   | 2548   |

## Reaction No. 39331



| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|----|---|-------|--------|-------------|---|-------|--------|
|----|---|-------|--------|-------------|---|-------|--------|

|   |    |      |                                  |                 |   |     |      |
|---|----|------|----------------------------------|-----------------|---|-----|------|
| 1 | 25 | 0.15 | DMSO:Water<br>80:20Vol<br>NaClO4 | lgK:66.1(0.1SD) | 5 | MGL | 2548 |
|---|----|------|----------------------------------|-----------------|---|-----|------|

| Reaction No. 39341                                  |    |       |                                  |                  |   |       |        |
|-----------------------------------------------------|----|-------|----------------------------------|------------------|---|-------|--------|
| 5<H+1> + [36]N12 + 2<Zn+2> = Zn+2(2)_H+1(5)_[36]N12 |    |       |                                  |                  |   |       |        |
| No                                                  | t  | I-Str | Medium                           | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                   | 25 | 0.15  | DMSO:Water<br>80:20Vol<br>NaClO4 | lgK:56.0(0.03SD) | 5 | MGL   | 2548   |

| Reaction No. 39368                            |    |       |        |               |   |       |        |
|-----------------------------------------------|----|-------|--------|---------------|---|-------|--------|
| H+1(3)_BisTren + SO4-2 = H+1(3)_BisTren_SO4-2 |    |       |        |               |   |       |        |
| No                                            | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                             | 25 | 0.1   | KCl    | lgK:2.25(3SF) | 6 | MGL   | 2459   |

| Reaction No. 39369                            |    |       |        |               |   |       |        |
|-----------------------------------------------|----|-------|--------|---------------|---|-------|--------|
| H+1(4)_BisTren + SO4-2 = H+1(4)_BisTren_SO4-2 |    |       |        |               |   |       |        |
| No                                            | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                             | 25 | 0.1   | KCl    | lgK:2.60(3SF) | 6 | MGL   | 2459   |

| Reaction No. 39373                            |    |       |        |               |   |       |        |
|-----------------------------------------------|----|-------|--------|---------------|---|-------|--------|
| H+1(5)_BisTren + SO4-2 = H+1(5)_BisTren_SO4-2 |    |       |        |               |   |       |        |
| No                                            | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                             | 25 | 0.1   | KCl    | lgK:2.72(3SF) | 6 | MGL   | 2459   |

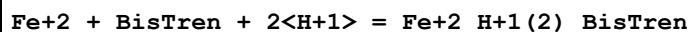
| Reaction No. 39375                            |    |       |        |               |   |       |        |
|-----------------------------------------------|----|-------|--------|---------------|---|-------|--------|
| H+1(6)_BisTren + SO4-2 = H+1(6)_BisTren_SO4-2 |    |       |        |               |   |       |        |
| No                                            | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                             | 25 | 0.1   | KCl    | lgK:3.63(3SF) | 6 | MGL   | 2459   |

| Reaction No. 39383                            |    |       |        |               |   |       |        |
|-----------------------------------------------|----|-------|--------|---------------|---|-------|--------|
| H+1(2)_BisTren + SO4-2 = H+1(2)_BisTren_SO4-2 |    |       |        |               |   |       |        |
| No                                            | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                             | 25 | 0.1   | KCl    | lgK:1.78(3SF) | 6 | MGL   | 2459   |

| Reaction No. 39391                                |   |       |        |             |   |       |        |
|---------------------------------------------------|---|-------|--------|-------------|---|-------|--------|
| 2<Fe+2> + BisTren = Fe+2(2)_H+1(-1)_BisTren + H+1 |   |       |        |             |   |       |        |
| No                                                | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |

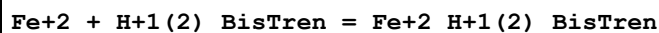
|   |    |     |     |                |   |     |      |
|---|----|-----|-----|----------------|---|-----|------|
| 1 | 25 | 0.1 | KCl | lgK:3.62 (3SF) | 6 | MGL | 2459 |
|---|----|-----|-----|----------------|---|-----|------|

## Reaction No. 39392



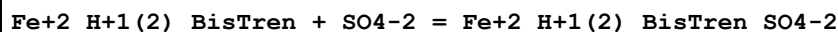
| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.1   | KCl    | lgK:23.22 (4SF) | 6 | MGL   | 2459   |

## Reaction No. 39393



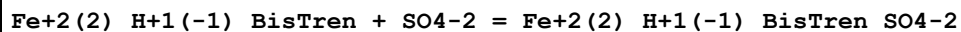
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | KCl    | lgK:4.14 (3SF) | 6 | MGL   | 2459   |

## Reaction No. 39395



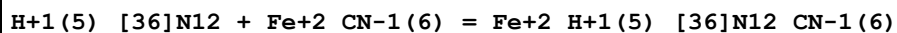
| No | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|--------|---------------|---|-------|--------|
| 1  | 25 | 0.1   | KCl    | lgK:1.6 (2SF) | 6 | MGL   | 2459   |

## Reaction No. 39396



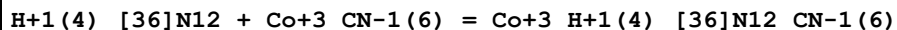
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | KCl    | lgK:2.64 (3SF) | 6 | MGL   | 2459   |

## Reaction No. 39804



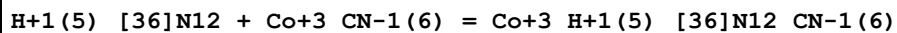
| No | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|--------|-------------------|---|-------|--------|
| 1  | 25 | 0.15  | NaClO4 | lgK:4.57 (0.02SD) | 5 | MGL   | 1782   |

## Reaction No. 39805



| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 0.15  | NaClO4 | lgK:3.2 (0.05SD) | 5 | MGL   | 1782   |

## Reaction No. 39806



| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 0.15  | NaClO4 | lgK:3.6 (0.05SD) | 5 | MGL   | 1782   |

## Reaction No. 39807

| H+1(6)_[36]N12 + Co+3_CN-1(6) = Co+3_H+1(6)_[36]N12_CN-1(6) |    |       |        |                 |   |       |        |
|-------------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| No                                                          | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                           | 25 | 0.15  | NaClO4 | lgK:3.8(0.04SD) | 5 | MGL   | 1782   |

| Reaction No. 39808                                          |    |       |        |                 |   |       |        |
|-------------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| H+1(7)_[36]N12 + Co+3_CN-1(6) = Co+3_H+1(7)_[36]N12_CN-1(6) |    |       |        |                 |   |       |        |
| No                                                          | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                           | 25 | 0.15  | NaClO4 | lgK:3.9(0.07SD) | 5 | MGL   | 1782   |

| Reaction No. 39809                                          |    |       |        |                 |   |       |        |
|-------------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| H+1(8)_[36]N12 + Co+3_CN-1(6) = Co+3_H+1(8)_[36]N12_CN-1(6) |    |       |        |                 |   |       |        |
| No                                                          | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                           | 25 | 0.15  | NaClO4 | lgK:4.2(0.07SD) | 5 | MGL   | 1782   |

| Reaction No. 39810                                          |    |       |        |                 |   |       |        |
|-------------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| H+1(9)_[36]N12 + Co+3_CN-1(6) = Co+3_H+1(9)_[36]N12_CN-1(6) |    |       |        |                 |   |       |        |
| No                                                          | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                           | 25 | 0.15  | NaClO4 | lgK:4.4(0.07SD) | 5 | MGL   | 1782   |

| Reaction No. 39811                                            |    |       |        |                 |   |       |        |
|---------------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| H+1(10)_[36]N12 + Co+3_CN-1(6) = Co+3_H+1(10)_[36]N12_CN-1(6) |    |       |        |                 |   |       |        |
| No                                                            | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                             | 25 | 0.15  | NaClO4 | lgK:4.4(0.09SD) | 5 | MGL   | 1782   |

| Reaction No. 39812                                            |    |       |        |                 |   |       |        |
|---------------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| H+1(11)_[36]N12 + Co+3_CN-1(6) = Co+3_H+1(11)_[36]N12_CN-1(6) |    |       |        |                 |   |       |        |
| No                                                            | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                             | 25 | 0.15  | NaClO4 | lgK:5.1(0.06SD) | 5 | MGL   | 1782   |

| Reaction No. 39813                                          |    |       |        |                  |   |       |        |
|-------------------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| H+1(6)_[36]N12 + Fe+2_CN-1(6) = Fe+2_H+1(6)_[36]N12_CN-1(6) |    |       |        |                  |   |       |        |
| No                                                          | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                           | 25 | 0.15  | NaClO4 | lgK:5.16(0.02SD) | 5 | MGL   | 1782   |

| Reaction No. 39814                                          |    |       |        |                 |   |       |        |
|-------------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| H+1(7)_[36]N12 + Fe+2_CN-1(6) = Fe+2_H+1(7)_[36]N12_CN-1(6) |    |       |        |                 |   |       |        |
| No                                                          | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                           | 25 | 0.15  | NaClO4 | lgK:6.0(0.05SD) | 5 | MGL   | 1782   |

| Reaction No. 39815                                                                                                              |    |       |        |                 |   |       |        |
|---------------------------------------------------------------------------------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| $\text{H+1(8)}_{[36]\text{N12}} + \text{Fe+2}_{\text{CN-1(6)}} = \text{Fe+2}_{\text{H+1(8)}}_{[36]\text{N12}_{\text{CN-1(6)}}}$ |    |       |        |                 |   |       |        |
| No                                                                                                                              | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                                                                               | 25 | 0.15  | NaClO4 | lgK:7.0(0.05SD) | 5 | MGL   | 1782   |

| Reaction No. 39816                                                                                                              |    |       |        |                 |   |       |        |
|---------------------------------------------------------------------------------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| $\text{H+1(9)}_{[36]\text{N12}} + \text{Fe+2}_{\text{CN-1(6)}} = \text{Fe+2}_{\text{H+1(9)}}_{[36]\text{N12}_{\text{CN-1(6)}}}$ |    |       |        |                 |   |       |        |
| No                                                                                                                              | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                                                                               | 25 | 0.15  | NaClO4 | lgK:7.5(0.06SD) | 5 | MGL   | 1782   |

| Reaction No. 41235                                                                                                            |    |       |               |              |   |       |        |
|-------------------------------------------------------------------------------------------------------------------------------|----|-------|---------------|--------------|---|-------|--------|
| $\text{Co+3}_{\text{CN-1(6)}} + [32]\text{N8} + 8\text{<H+1>} = \text{Co+3}_{\text{H+1(8)}}_{[32]\text{N8}_{\text{CN-1(6)}}}$ |    |       |               |              |   |       |        |
| No                                                                                                                            | t  | I-Str | Medium        | Value (Dev)  | W | Tech. | Ref. # |
| 1                                                                                                                             | 23 | 0.1   | Me4NC1        | lgK:6.0(2SF) | 0 | MGL   | 4382   |
| 2                                                                                                                             | 25 | 0     | Inf. Dilution | lgK:6.2(2SF) | 1 | ESC   |        |

| Reaction No. 41239                                                                                                            |    |       |               |              |   |       |        |
|-------------------------------------------------------------------------------------------------------------------------------|----|-------|---------------|--------------|---|-------|--------|
| $\text{Fe+2}_{\text{CN-1(6)}} + [32]\text{N8} + 8\text{<H+1>} = \text{Fe+2}_{\text{H+1(8)}}_{[32]\text{N8}_{\text{CN-1(6)}}}$ |    |       |               |              |   |       |        |
| No                                                                                                                            | t  | I-Str | Medium        | Value (Dev)  | W | Tech. | Ref. # |
| 1                                                                                                                             | 23 | 0.1   | Me4NC1        | lgK:8.9(2SF) | 5 | MGL   | 4382   |
| 2                                                                                                                             | 25 | 0     | Inf. Dilution | lgK:9.1(2SF) | 1 | ESC   |        |

| Reaction No. 41687                                                          |    |       |                          |                |   |       |        |
|-----------------------------------------------------------------------------|----|-------|--------------------------|----------------|---|-------|--------|
| $[24]\text{O8:DiBenzo} + \text{Hg+2} = \text{Hg+2}_{[24]\text{O8:DiBenzo}}$ |    |       |                          |                |   |       |        |
| No                                                                          | t  | I-Str | Medium                   | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                           | 25 | 0.1   | MeCN<br>Et4NC1O4         | lgK:2.(1SF)    | 3 | MPL   | 4472   |
| 2                                                                           | 25 | 0.1   | Nitrobenzene<br>Et4NC1O4 | lgK:3.2(0.1SD) | 5 | MPL   | 4472   |

| Reaction No. 42750                                                                                    |    |       |        |               |   |       |        |
|-------------------------------------------------------------------------------------------------------|----|-------|--------|---------------|---|-------|--------|
| $5\text{<H+1>} + [32]\text{N8} + 2\text{<5AMP-2>} = \text{H+1(5)}_{[32]\text{N8}_{5\text{AMP-2(2)}}}$ |    |       |        |               |   |       |        |
| No                                                                                                    | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                                                                     | 25 | 0.1   | Me4NC1 | lgK:5.00(3SF) | 5 | MMR   | 4353   |

| Reaction No. 42751                                                                                    |   |       |        |             |   |       |        |
|-------------------------------------------------------------------------------------------------------|---|-------|--------|-------------|---|-------|--------|
| $6\text{<H+1>} + [32]\text{N8} + 2\text{<5AMP-2>} = \text{H+1(6)}_{[32]\text{N8}_{5\text{AMP-2(2)}}}$ |   |       |        |             |   |       |        |
| No                                                                                                    | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |



|   |    |     |        |                |   |     |      |
|---|----|-----|--------|----------------|---|-----|------|
| 1 | 25 | 0.1 | Me4NC1 | lgK:5.55 (3SF) | 5 | MMR | 4353 |
|---|----|-----|--------|----------------|---|-----|------|

## Reaction No. 42752



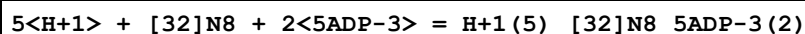
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | Me4NC1 | lgK:5.90 (3SF) | 5 | MMR   | 4353   |

## Reaction No. 42753



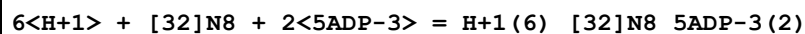
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | Me4NC1 | lgK:7.20 (3SF) | 5 | MMR   | 4353   |

## Reaction No. 42754



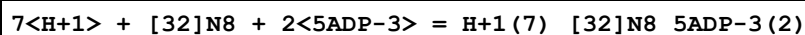
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | Me4NC1 | lgK:7.00 (3SF) | 5 | MMR   | 4353   |

## Reaction No. 42755



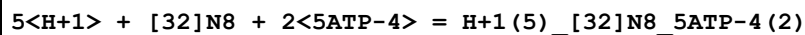
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | Me4NC1 | lgK:8.10 (3SF) | 5 | MMR   | 4353   |

## Reaction No. 42756



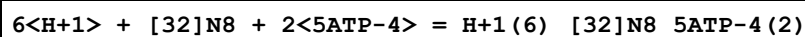
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | Me4NC1 | lgK:9.15 (3SF) | 5 | MMR   | 4353   |

## Reaction No. 42757



| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | Me4NC1 | lgK:8.00 (3SF) | 5 | MMR   | 4353   |

## Reaction No. 42758



| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | Me4NC1 | lgK:9.95 (3SF) | 5 | MMR   | 4353   |

## Reaction No. 42759

| 7<H+1> + [32]N8 + 2<5ATP-4> = H+1(7)_[32]N8_5ATP-4(2) |    |       |        |                |   |       |        |
|-------------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| No                                                    | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                     | 25 | 0.1   | Me4NC1 | lgK:11.50(4SF) | 5 | MMR   | 4353   |

| Reaction No. 42896                                    |    |       |        |                |   |       |        |
|-------------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| 8<H+1> + [32]N8 + 2<5ADP-3> = H+1(8)_[32]N8_5ADP-3(2) |    |       |        |                |   |       |        |
| No                                                    | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                     | 25 | 0.1   | Me4NC1 | lgK:10.20(4SF) | 5 | MMR   | 4353   |

| Reaction No. 42897                                    |    |       |        |                |   |       |        |
|-------------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| 8<H+1> + [32]N8 + 2<5ATP-4> = H+1(8)_[32]N8_5ATP-4(2) |    |       |        |                |   |       |        |
| No                                                    | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                     | 25 | 0.1   | Me4NC1 | lgK:12.80(4SF) | 5 | MMR   | 4353   |

| Reaction No. 43817                                |    |       |        |                |   |       |        |
|---------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| H+1 + SemiGlycresolRed-3 = H+1_SemiGlycresolRed-3 |    |       |        |                |   |       |        |
| No                                                | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                 | 25 | 0.1   | KNO3   | lgK:10.07(4SF) | 5 | MSP   | 2404   |

| Reaction No. 43818                                       |    |       |        |                |   |       |        |
|----------------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| H+1_SemiGlycresolRed-3 + H+1 = H+1(2)_SemiGlycresolRed-3 |    |       |        |                |   |       |        |
| No                                                       | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                        | 25 | 0.1   | KNO3   | lgK:7.44(3SF)w | 5 | MPH   | 2404   |

| Reaction No. 43819                                          |    |       |        |                |   |       |        |
|-------------------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| H+1(2)_SemiGlycresolRed-3 + H+1 = H+1(3)_SemiGlycresolRed-3 |    |       |        |                |   |       |        |
| No                                                          | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                           | 25 | 0.1   | KNO3   | lgK:2.47(3SF)w | 5 | MPH   | 2404   |

| Reaction No. 43820                                          |    |       |        |                |   |       |        |
|-------------------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| H+1(3)_SemiGlycresolRed-3 + H+1 = H+1(4)_SemiGlycresolRed-3 |    |       |        |                |   |       |        |
| No                                                          | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                           | 25 | 0.1   | KNO3   | lgK:-0.27(2SF) | 5 | MSP   | 2404   |

| Reaction No. 43821                                          |    |       |        |                |   |       |        |
|-------------------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| H+1(4)_SemiGlycresolRed-3 + H+1 = H+1(5)_SemiGlycresolRed-3 |    |       |        |                |   |       |        |
| No                                                          | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                           | 25 | 0.1   | KNO3   | lgK:-1.76(3SF) | 5 | MSP   | 2404   |

| Reaction No. 43822                                  |    |       |        |               |   |       |        |
|-----------------------------------------------------|----|-------|--------|---------------|---|-------|--------|
| SemiGlycresolRed-3 + Mn+2 = Mn+2_SemiGlycresolRed-3 |    |       |        |               |   |       |        |
| No                                                  | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                   | 25 | 0.1   | KNO3   | lgK:4.6(2SF)w | 5 | MPH   | 2404   |

| Reaction No. 43823                                                        |    |       |        |               |   |       |        |
|---------------------------------------------------------------------------|----|-------|--------|---------------|---|-------|--------|
| Mn+2_SemiGlycresolRed-3 + SemiGlycresolRed-3 = Mn+2_SemiGlycresolRed-3(2) |    |       |        |               |   |       |        |
| No                                                                        | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                                         | 25 | 0.1   | KNO3   | lgK:3.1(2SF)w | 5 | MPH   | 2404   |

| Reaction No. 43824                                  |    |       |        |               |   |       |        |
|-----------------------------------------------------|----|-------|--------|---------------|---|-------|--------|
| SemiGlycresolRed-3 + Co+2 = Co+2_SemiGlycresolRed-3 |    |       |        |               |   |       |        |
| No                                                  | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                   | 25 | 0.1   | KNO3   | lgK:7.6(2SF)w | 5 | MPH   | 2404   |

| Reaction No. 43825                                                        |    |       |        |               |   |       |        |
|---------------------------------------------------------------------------|----|-------|--------|---------------|---|-------|--------|
| Co+2_SemiGlycresolRed-3 + SemiGlycresolRed-3 = Co+2_SemiGlycresolRed-3(2) |    |       |        |               |   |       |        |
| No                                                                        | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                                         | 25 | 0.1   | KNO3   | lgK:5.4(2SF)w | 5 | MPH   | 2404   |

| Reaction No. 43826                                  |    |       |        |               |   |       |        |
|-----------------------------------------------------|----|-------|--------|---------------|---|-------|--------|
| SemiGlycresolRed-3 + Ni+2 = Ni+2_SemiGlycresolRed-3 |    |       |        |               |   |       |        |
| No                                                  | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                   | 25 | 0.1   | KNO3   | lgK:8.8(2SF)w | 5 | MPH   | 2404   |

| Reaction No. 43827                                                        |    |       |        |               |   |       |        |
|---------------------------------------------------------------------------|----|-------|--------|---------------|---|-------|--------|
| Ni+2_SemiGlycresolRed-3 + SemiGlycresolRed-3 = Ni+2_SemiGlycresolRed-3(2) |    |       |        |               |   |       |        |
| No                                                                        | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                                         | 25 | 0.1   | KNO3   | lgK:6.1(2SF)w | 5 | MPH   | 2404   |

| Reaction No. 43828                                            |    |       |        |              |   |       |        |
|---------------------------------------------------------------|----|-------|--------|--------------|---|-------|--------|
| Ni+2_SemiGlycresolRed-3 + OH-1 = Ni+2_OH-1_SemiGlycresolRed-3 |    |       |        |              |   |       |        |
| No                                                            | t  | I-Str | Medium | Value (Dev)  | W | Tech. | Ref. # |
| 1                                                             | 25 | 0.1   | KNO3   | lgK:3.8(2SF) | 5 | MSP   | 2404   |

| Reaction No. 43829                                  |  |  |  |  |  |  |  |
|-----------------------------------------------------|--|--|--|--|--|--|--|
| SemiGlycresolRed-3 + Cu+2 = Cu+2_SemiGlycresolRed-3 |  |  |  |  |  |  |  |

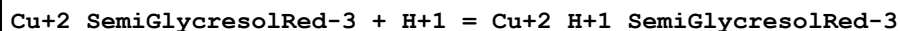
| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 0.1   | KNO3   | lgK:12.8 (3SF) w | 5 | MPH   | 2404   |

## Reaction No. 43830



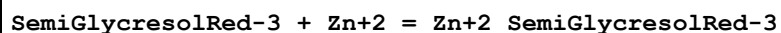
| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.1   | KNO3   | lgK:5.2 (2SF) w | 5 | MPH   | 2404   |

## Reaction No. 43831



| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.1   | KNO3   | lgK:5.3 (2SF) w | 5 | MPH   | 2404   |

## Reaction No. 43832



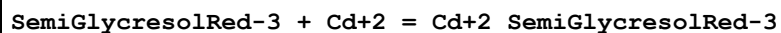
| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.1   | KNO3   | lgK:7.6 (2SF) w | 5 | MPH   | 2404   |

## Reaction No. 43833



| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.1   | KNO3   | lgK:5.7 (2SF) w | 5 | MPH   | 2404   |

## Reaction No. 43834



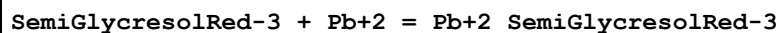
| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.1   | KNO3   | lgK:6.2 (2SF) w | 5 | MPH   | 2404   |

## Reaction No. 43835



| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.1   | KNO3   | lgK:4.3 (2SF) w | 5 | MPH   | 2404   |

## Reaction No. 43836



| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.1   | KNO3   | lgK:9.1 (2SF) w | 5 | MPH   | 2404   |

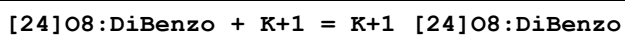
| Reaction No. 43837                                                        |    |       |        |                |   |       |        |
|---------------------------------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| Pb+2_SemiGlycresolRed-3 + SemiGlycresolRed-3 = Pb+2_SemiGlycresolRed-3(2) |    |       |        |                |   |       |        |
| No                                                                        | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                         | 25 | 0.1   | KNO3   | lgK:3.4 (2SF)w | 5 | MPH   | 2404   |

| Reaction No. 43838                                            |    |       |        |               |   |       |        |
|---------------------------------------------------------------|----|-------|--------|---------------|---|-------|--------|
| Pb+2_SemiGlycresolRed-3 + OH-1 = Pb+2_OH-1_SemiGlycresolRed-3 |    |       |        |               |   |       |        |
| No                                                            | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                             | 25 | 0.1   | KNO3   | lgK:4.0 (2SF) | 5 | MSP   | 2404   |

| Reaction No. 45660                          |     |       |                            |                   |   |       |        |
|---------------------------------------------|-----|-------|----------------------------|-------------------|---|-------|--------|
| [24]O8:DiBenzo + Cs+1 = Cs+1_[24]O8:DiBenzo |     |       |                            |                   |   |       |        |
| No                                          | t   | I-Str | Medium                     | Value (Dev)       | W | Tech. | Ref. # |
| 1                                           | 0   |       | DMF                        | lgK:2.65 (0.02SD) | 4 | MMR   | 3845   |
| 2                                           | 10  |       | DMF                        | lgK:2.5 (0.03SD)  | 4 | MMR   | 3845   |
| 3                                           | 20  |       | DMF                        | lgK:2.32 (0.01SD) | 4 | MMR   | 3845   |
| 4                                           | 27  | 0     | DMF<br>Inf. Dilution       | lgK:2.21 (0.03SD) | 5 | MMR   | 6236   |
| 5                                           | 30  |       | DMF                        | lgK:2.2 (0.03SD)  | 4 | MMR   | 3845   |
| 6                                           | 40  |       | DMF                        | lgK:2.02 (0.01SD) | 4 | MMR   | 3845   |
| 7                                           | 50  |       | DMF                        | lgK:1.89 (0.01SD) | 4 | MMR   | 3845   |
| 8                                           | 0   |       | DMF:MeCN<br>22.69:77.31Vol | lgK:3.6 (0.05SD)  | 4 | MMR   | 3845   |
| 9                                           | 10  |       | DMF:MeCN<br>22.69:77.31Vol | lgK:3.35 (0.02SD) | 4 | MMR   | 3845   |
| 10                                          | 20  |       | DMF:MeCN<br>22.69:77.31Vol | lgK:3.18 (0.01SD) | 4 | MMR   | 3845   |
| 11                                          | 30  |       | DMF:MeCN<br>22.69:77.31Vol | lgK:3.03 (0.01SD) | 4 | MMR   | 3845   |
| 12                                          | 40  |       | DMF:MeCN<br>22.69:77.31Vol | lgK:2.85 (0.02SD) | 4 | MMR   | 3845   |
| 13                                          | 50  |       | DMF:MeCN<br>22.69:77.31Vol | lgK:2.68 (0.02SD) | 4 | MMR   | 3845   |
| 14                                          | -10 |       | DMF:MeCN<br>61.5:38.5Vol   | lgK:3.11 (0.02SD) | 4 | MMR   | 3845   |
| 15                                          | 0   |       | DMF:MeCN<br>61.5:38.5Vol   | lgK:2.98 (0.02SD) | 4 | MMR   | 3845   |
| 16                                          | 10  |       | DMF:MeCN<br>61.5:38.5Vol   | lgK:2.78 (0.02SD) | 4 | MMR   | 3845   |
| 17                                          | 20  |       | DMF:MeCN<br>61.5:38.5Vol   | lgK:2.54 (0.02SD) | 4 | MMR   | 3845   |

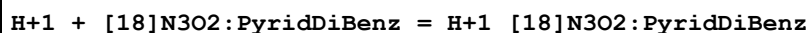
|    |    |     |                                       |                   |   |     |      |
|----|----|-----|---------------------------------------|-------------------|---|-----|------|
| 18 | 30 |     | DMF:MeCN<br>61.5:38.5Vol              | lgK:2.4 (0.03SD)  | 4 | MMR | 3845 |
| 19 | 40 |     | DMF:MeCN<br>61.5:38.5Vol              | lgK:2.3 (0.03SD)  | 4 | MMR | 3845 |
| 20 | 5  |     | MeCN                                  | lgK:4.5 (0.05SD)  | 4 | MMR | 3845 |
| 21 | 30 |     | MeCN                                  | lgK:3.9 (0.07SD)  | 4 | MMR | 3845 |
| 22 | 40 |     | MeCN                                  | lgK:3.8 (0.03SD)  | 4 | MMR | 3845 |
| 23 | 50 |     | MeCN                                  | lgK:3.6 (0.05SD)  | 4 | MMR | 3845 |
| 24 | 60 |     | MeCN                                  | lgK:3.5 (0.04SD)  | 4 | MMR | 3845 |
| 25 | 75 |     | MeCN                                  | lgK:3.2 (0.03SD)  | 4 | MMR | 3845 |
| 26 | 25 |     | Methanol                              | lgK:3.8 (0.08SD)  | 4 | MIS | 6509 |
| 27 | 25 | 0.1 | Methanol:Water<br>70:30Wgt<br>Unknown | lgK:2.48 (0.01SD) | 5 | MCL | 4522 |

## Reaction No. 45661



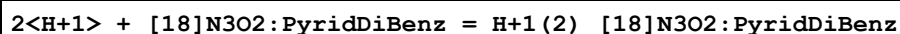
| No | t  | I-Str | Medium                                | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|---------------------------------------|-------------------|---|-------|--------|
| 1  | 25 |       | Methanol                              | lgK:3.5 (0.04SD)  | 4 | MIS   | 6509   |
| 2  | 25 | 0.1   | Methanol:Water<br>70:30Wgt<br>Unknown | lgK:2.42 (0.01SD) | 5 | MCL   | 4522   |

## Reaction No. 58611



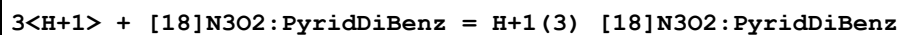
| No | t  | I-Str | Medium                                | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------------------------------|-----------------|---|-------|--------|
| 1  | 25 | 0.1   | Water:Methanol<br>5:95Vol<br>Et4NC1O4 | lgK:10.40 (4SF) | 5 | MGL   | 5371   |

## Reaction No. 58612



| No | t  | I-Str | Medium                                | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------------------------------|-----------------|---|-------|--------|
| 1  | 25 | 0.1   | Water:Methanol<br>5:95Vol<br>Et4NC1O4 | lgK:17.57 (4SF) | 4 | MGL   | 5371   |

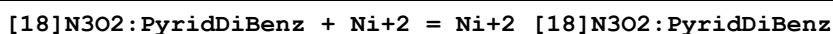
## Reaction No. 58613



| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|----|---|-------|--------|-------------|---|-------|--------|
|----|---|-------|--------|-------------|---|-------|--------|

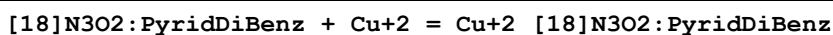
|   |    |     |                                       |                |   |     |      |
|---|----|-----|---------------------------------------|----------------|---|-----|------|
| 1 | 25 | 0.1 | Water:Methanol<br>5:95Vol<br>Et4NC1O4 | lgK:19.86(4SF) | 3 | MGL | 5371 |
|---|----|-----|---------------------------------------|----------------|---|-----|------|

## Reaction No. 58614



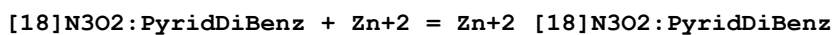
| No | t  | I-Str | Medium                                | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|---------------------------------------|---------------|---|-------|--------|
| 1  | 25 | 0.1   | Water:Methanol<br>5:95Vol<br>Et4NC1O4 | lgK:5.15(3SF) | 4 | MGL   | 5371   |

## Reaction No. 58615



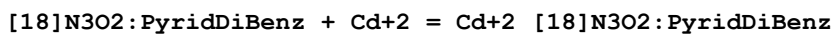
| No | t  | I-Str | Medium                                | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|---------------------------------------|---------------|---|-------|--------|
| 1  | 25 | 0.1   | Water:Methanol<br>5:95Vol<br>Et4NC1O4 | lgK:8.72(3SF) | 4 | MGL   | 5371   |
| 2  | 25 | 0.1   | Water:Methanol<br>5:95Vol<br>Et4NC1O4 | dH:-6.41(3SF) | 4 | MCL   | 5371   |

## Reaction No. 58660



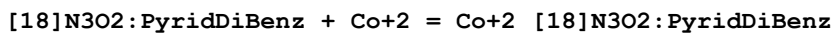
| No | t  | I-Str | Medium                                | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|---------------------------------------|------------------|---|-------|--------|
| 1  | 25 | 0.1   | Methanol:Water<br>95:5Vol<br>Et4NC1O4 | lgK:5.75(0.05SD) | 5 | MGL   | 4869   |

## Reaction No. 58661



| No | t  | I-Str | Medium                                | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|---------------------------------------|------------------|---|-------|--------|
| 1  | 25 | 0.1   | Methanol:Water<br>95:5Vol<br>Et4NC1O4 | lgK:4.19(0.05SD) | 5 | MGL   | 4869   |

## Reaction No. 58662



| No | t  | I-Str | Medium                                | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|---------------------------------------|---------------|---|-------|--------|
| 1  | 25 | 0.1   | Methanol:Water<br>95:5Vol<br>Et4NC1O4 | lgK:4.36(3SF) | 5 | MGL   | 4870   |

## Reaction No. 58663

| [18]N3O2:PyridDiBenz + Mn+2 = Mn+2_[18]N3O2:PyridDiBenz |    |       |                                       |              |   |       |        |
|---------------------------------------------------------|----|-------|---------------------------------------|--------------|---|-------|--------|
| No                                                      | t  | I-Str | Medium                                | Value (Dev)  | W | Tech. | Ref. # |
| 1                                                       | 25 | 0.1   | Methanol:Water<br>95:5Vol<br>Et4NC1O4 | lgK:3.9(2SF) | 3 | MGL   | 4870   |

| Reaction No. 59074                          |    |       |                                       |                  |   |       |        |
|---------------------------------------------|----|-------|---------------------------------------|------------------|---|-------|--------|
| [24]O8:DiBenzo + Na+1 = Na+1_[24]O8:DiBenzo |    |       |                                       |                  |   |       |        |
| No                                          | t  | I-Str | Medium                                | Value (Dev)      | W | Tech. | Ref. # |
| 1                                           | 25 | 0.1   | Methanol:Water<br>70:30Wgt<br>Unknown | lgK:1.54(0.01SD) | 5 | MCL   | 4522   |

| Reaction No. 59075                          |    |       |                                       |                  |   |       |        |
|---------------------------------------------|----|-------|---------------------------------------|------------------|---|-------|--------|
| [24]O8:DiBenzo + Rb+1 = Rb+1_[24]O8:DiBenzo |    |       |                                       |                  |   |       |        |
| No                                          | t  | I-Str | Medium                                | Value (Dev)      | W | Tech. | Ref. # |
| 1                                           | 27 | 0     | DMF<br>Inf. Dilution                  | lgK:1.74(0.03SD) | 5 | MMR   | 6236   |
| 2                                           | 25 | 0.1   | Methanol:Water<br>70:30Wgt<br>Unknown | lgK:2.55(0.01SD) | 5 | MCL   | 4522   |

| Reaction No. 59111                             |    |       |        |               |   |       |        |
|------------------------------------------------|----|-------|--------|---------------|---|-------|--------|
| Cu+2 + BisTren + H2O = Cu+2_BisTren_OH-1 + H+1 |    |       |        |               |   |       |        |
| No                                             | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                              | 25 | 0.1   | KCl    | lgK:6.17(3SF) | 1 | MGL   | 1545   |
| 2                                              | 25 | 0.1   | NaClO4 | lgK:6.92(3SF) | 1 | MGL   | 1545   |

| Reaction No. 59112                          |    |       |        |              |   |       |        |
|---------------------------------------------|----|-------|--------|--------------|---|-------|--------|
| Zn+2(2)_[36]N12 + H+1 = Zn+2(2)_H+1_[36]N12 |    |       |        |              |   |       |        |
| No                                          | t  | I-Str | Medium | Value (Dev)  | W | Tech. | Ref. # |
| 1                                           | 25 | 0.15  | NaClO4 | lgK:6.6(2SF) | 5 | MPT   | 4257   |

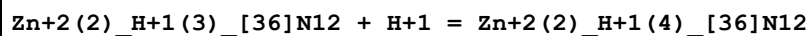
| Reaction No. 59113                                 |    |       |        |              |   |       |        |
|----------------------------------------------------|----|-------|--------|--------------|---|-------|--------|
| Zn+2(2)_H+1_[36]N12 + H+1 = Zn+2(2)_H+1(2)_[36]N12 |    |       |        |              |   |       |        |
| No                                                 | t  | I-Str | Medium | Value (Dev)  | W | Tech. | Ref. # |
| 1                                                  | 25 | 0.15  | NaClO4 | lgK:6.3(2SF) | 5 | MPT   | 4257   |

| Reaction No. 59114                                    |  |  |  |  |  |  |  |
|-------------------------------------------------------|--|--|--|--|--|--|--|
| Zn+2(2)_H+1(2)_[36]N12 + H+1 = Zn+2(2)_H+1(3)_[36]N12 |  |  |  |  |  |  |  |



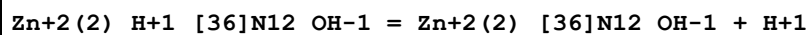
| No | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|--------|---------------|---|-------|--------|
| 1  | 25 | 0.15  | NaClO4 | lgK:5.7 (2SF) | 5 | MPT   | 4257   |

## Reaction No. 59115



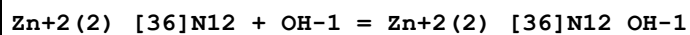
| No | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|--------|---------------|---|-------|--------|
| 1  | 25 | 0.15  | NaClO4 | lgK:5.2 (2SF) | 5 | MPT   | 4257   |

## Reaction No. 59116



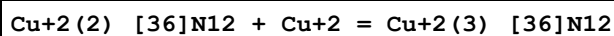
| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.15  | NaClO4 | lgK:-10.1 (3SF) | 5 | MPT   | 4257   |

## Reaction No. 59117



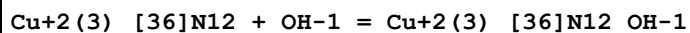
| No | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|--------|---------------|---|-------|--------|
| 1  | 25 | 0.15  | NaClO4 | lgK:3.5 (2SF) | 5 | MPT   | 4257   |

## Reaction No. 59124



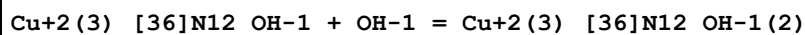
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.15  | NaClO4 | lgK:12.2 (3SF) | 5 | MGL   | 1341   |

## Reaction No. 59125



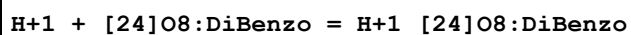
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.15  | NaClO4 | lgK:4.83 (3SF) | 5 | MGL   | 1341   |

## Reaction No. 59126



| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.15  | NaClO4 | lgK:2.82 (3SF) | 5 | MGL   | 1341   |

## Reaction No. 69149



| No | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|--------|---------------|---|-------|--------|
| 1  | 25 |       | MeCN   | lgK:3.2 (2SF) | 4 | MCN   | 4967   |

| Reaction No. 76497                                         |    |       |               |                   |   |       |        |
|------------------------------------------------------------|----|-------|---------------|-------------------|---|-------|--------|
| $\text{Me4N+1\_Ph4B-1(s)} = \text{Me4N+1} + \text{Ph4B-1}$ |    |       |               |                   |   |       |        |
| No                                                         | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                          | 25 | 0     | Inf. Dilution | dG:61.92 (4SF) kJ | 7 | CRV   | 22677  |

| Reaction No. 80964                                     |    |       |               |                 |   |       |        |
|--------------------------------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| $\text{Ag+1\_Ph4B-1(s)} = \text{Ag+1} + \text{Ph4B-1}$ |    |       |               |                 |   |       |        |
| No                                                     | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                      | 25 | 0     | Inf. Dilution | lgK:-17.2 (3SF) | 2 | MIX   | 32205  |

| Reaction No. 80965                                           |    |       |               |                 |   |       |        |
|--------------------------------------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| $\text{Ph4As+1\_ClO4-1(s)} = \text{Ph4As+1} + \text{ClO4-1}$ |    |       |               |                 |   |       |        |
| No                                                           | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                            | 25 | 0     | Inf. Dilution | lgK:-8.40 (3SF) | 4 | MSL   | 32204  |

| Reaction No. 80966                                           |    |       |               |                 |   |       |        |
|--------------------------------------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| $\text{Ph4As+1\_ReO4-1(s)} = \text{Ph4As+1} + \text{ReO4-1}$ |    |       |               |                 |   |       |        |
| No                                                           | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                            | 25 | 0     | Inf. Dilution | lgK:-8.49 (3SF) | 4 | MSL   | 32204  |

| Reaction No. 80967                                           |    |       |               |                 |   |       |        |
|--------------------------------------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| $\text{Ph4As+1\_MnO4-1(s)} = \text{Ph4As+1} + \text{MnO4-1}$ |    |       |               |                 |   |       |        |
| No                                                           | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                            | 25 | 0     | Inf. Dilution | lgK:-8.09 (3SF) | 4 | MSL   | 32204  |

| Reaction No. 80968                                           |    |       |               |                 |   |       |        |
|--------------------------------------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| $\text{Ph4As+1\_Ph4B-1(s)} = \text{Ph4As+1} + \text{Ph4B-1}$ |    |       |               |                 |   |       |        |
| No                                                           | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                            | 25 | 0     | Inf. Dilution | lgK:-8.30 (3SF) | 4 | MSL   | 32204  |

| Reaction No. 80969                                                         |    |       |               |                 |   |       |        |
|----------------------------------------------------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| $\text{Co+2\_Ph4As+1\_Cl-1(3)(s)} = \text{Ph4As+1} + \text{Co+2\_Cl-1(3)}$ |    |       |               |                 |   |       |        |
| No                                                                         | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                          | 25 | 0     | Inf. Dilution | lgK:-9.17 (3SF) | 4 | MSL   | 32204  |

| Reaction No. 80970                                                   |   |       |        |             |   |       |        |
|----------------------------------------------------------------------|---|-------|--------|-------------|---|-------|--------|
| $\text{Ph4As+1(2)\_Cr2O7-2(s)} = 2<\text{Ph4As+1}> + \text{Cr2O7-2}$ |   |       |        |             |   |       |        |
| No                                                                   | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |

|   |    |   |               |                |   |     |       |
|---|----|---|---------------|----------------|---|-----|-------|
| 1 | 25 | 0 | Inf. Dilution | lgK:-9.19(3SF) | 4 | MSL | 32204 |
|---|----|---|---------------|----------------|---|-----|-------|

| Reaction No. 80971                                      |    |       |               |                 |   |       |        |
|---------------------------------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| Zn+2_Ph4As+1(2)_Cl-1(4)_(s) = 2<Ph4As+1> + Zn+2_Cl-1(4) |    |       |               |                 |   |       |        |
| No                                                      | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                       | 25 | 0     | Inf. Dilution | lgK:-12.49(4SF) | 4 | MSL   | 32204  |

### Data Assessment

| Weight | Assessment                                    |
|--------|-----------------------------------------------|
| 9      | highest accuracy                              |
| 8      | multi-determined; excellent dependability     |
| 7      | trustworthy; outstanding single determination |
| 6      | better than average reliability               |
| 5      | average single determination                  |
| 4      | some doubt                                    |
| 3      | poorly known                                  |
| 2      | guess to achieve consistency (or just as bad) |
| 1      | no information but consistent                 |
| 0      | problematic/inconsistent                      |

### Techniques

| Abbrev | Method                          |
|--------|---------------------------------|
| CRV    | Critical review                 |
| EDH    | Debye-Hueckel corrections       |
| ELC    | Linear Combination of Reactions |
| ESC    | Standard conditions anchor      |
| MCL    | Calorimetry                     |
| MCN    | Conductivity                    |
| MGL    | Glass electrode                 |
| MIS    | Ion selective electrode         |
| MIX    | Ion exchange                    |

|     |                            |
|-----|----------------------------|
| MMR | Nuclear magnetic resonance |
| MPH | pH method, not specified   |
| MPL | Polarography               |
| MPT | Potentiometry              |
| MSL | Solubility                 |
| MSP | Spectroscopy               |

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